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Reinforcing bars modelling using a rod–solid interface element without the need for mesh compatibility

R. Durand [a,*], M.M. Farias [a], D.M. Pedroso [b], G. Meschke [c]

[a] Department of Civil and Environmental Engineering, University of Brasilia, Campus Darcy Ribeiro, Brasilia DF, CEP 70910-900, Brazil
[b] School of Civil Engineering, The University of Queensland, St Lucia QLD 4072, Australia
[c] Institute for Structural Mechanics, Department for Civil and Environmental Engineering Sciences, Ruhr University Bochum, Germany

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ABSTRACT

This paper presents a finite element methodology for modelling solid media with any number of straight or curved reinforcing bars (rebars). The formulation can handle rebars arbitrarily located within a 2D or 3D finite element mesh without the need for nodal compatibility. A new continuous rod–solid interface (or joint) element is developed to connect bulk elements to the rebars, including a method that directly considers the relative displacements. An arbitrary constitutive model can be used in conjunction with the proposed interface element to represent slip behaviour. In this regard, we also present a cyclic nonlinear constitutive model. Unlike existing embedded approaches, the proposed method allows for the direct application of boundary conditions at reinforcements' endpoints, including tip bearing. The formulation is verified by comparing with literature examples and analytical solutions. In all cases, excellent agreement is observed. The approach can be conveniently applied to the study of complex reinforced concrete structures or any solid medium where reinforcing bars improve its mechanical properties.

1. Introduction

Reinforcing bars are commonly used in Engineering to improve the mechanical strength of structural members. Examples of applications are reinforced concrete (RC), soil nailing, and tunnel excavation with the NATM technique, where bars are inserted while the ground is excavated. In Engineering, the finite element method (FEM) is the most popular numerical technique for solving continuum mechanical problems, thanks to its versatility in handling complex geometries and nonlinear constitutive laws; see, e.g. [1–5]. Nonetheless, the modelling of reinforcing bars in solids is still a difficult task with conventional FE codes. One problem is defining a proper interface model for linking the reinforcing bars (rebars) to 2D or 3D (bulk) elements. The other problem is the strict constraint regarding mesh topology.

A major difficulty in modelling reinforced solids is the meshing of one-dimensional elements inserted in solids in 2D or 3D so that the compatibility of nodes is preserved (Fig. 1(a)). This difficulty is aggravated when many bars need to be positioned in the solid, as in the case of reinforced concrete structures. Three methods are described in the literature to model the inclusion of such line elements in structures; herein, they are named *discrete*, *distributed* and *embedded* and are briefly reviewed as follows.

The discrete method represents the conventional finite element approach to deal with reinforced materials. Reinforcing bars are modelled by connecting bar elements directly to the nodes of solid elements (Fig. 1(a)) or using special elements such as springs to represent bondings. It is simple and has the advantage of representing different materials in a precise way. For instance, Smith and Su [6] and Zhang et al. [7] applied this method to simulate horizontal soil nails within structured 3D meshes. The discrete method's main disadvantage is that the finite element mesh design is highly dependent on the reinforcement layout. To circumvent this problem, El-Mezaini and Çitipitioğlu [8] developed elements with movable mid-side nodes to avoid nodal compatibility problems. More recently, Murcia-Delso and Benson Shing [9] modelled 3D RC specimens using 4-node interface elements that connect rod or beam elements to the edges of surrounding solid elements. In a variant of the discrete method, Manzoli et al. [10] simulated RC structures with the aid of 2D thin elements with high aspect ratio to simulate the concrete–steel contact behaviour.

In the distributed method, reinforcements are considered smeared over bulk elements at particular angles, and perfect bonding is implicitly assumed. The total equivalent stiffness of the resulting composite element is a compound of the stiffnesses of all elements, solid and

* Corresponding author.
E-mail addresses: durand@unb.br (R. Durand), muniz@unb.br (M.M. Farias), d.pedroso@uq.edu.au (D.M. Pedroso), guenther.meschke@rub.de (G. Meschke).

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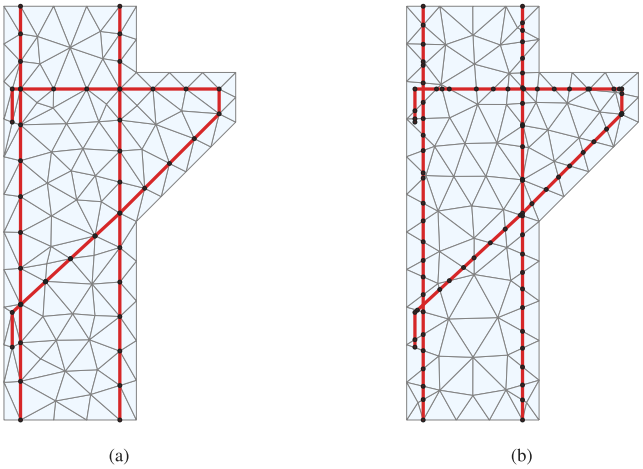


Fig. 1. Finite element meshes for a reinforced structural member: (a) discrete (conventional) method with compatibility between rod and bulk elements; (b) embedded method (incompatible mesh). Note: meshes drawn for illustration only since the situation indicated can only be accurately modelled in 3D.

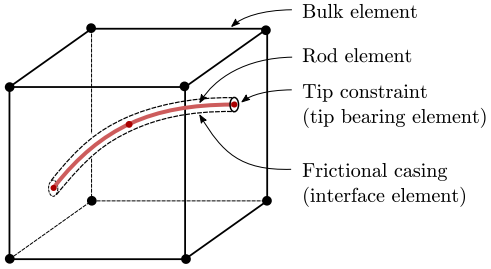


Fig. 2. Schematic illustration of the element types involved in the proposed method.

rebars. This type of formulation is helpful for the analysis of reinforced structures with densely and uniformly distributed reinforcement; however, it cannot determine stresses on rebars accurately. Examples of this method's applications are given by Pietruszczak and Winnicki [11], who simulated reinforced concrete with two orthogonal sets of steel rebars as a composite material.

In the embedded method [12–17], mesh generation problems are entirely avoided since the solid mesh is independent of the reinforcement layout. Therefore, reinforcing bars can cross any bulk element and can also be incompatible (Fig. 1(b)). The major advantage of this method is that a single pre-generated background mesh can be used to analyse a number of different reinforcement configurations for designing purposes. A pre-processing tool is required to compute the rebars positions and intersections with bulk elements. Usually, the stiffness of reinforcing bars are incorporated into the stiffness of crossed elements. Strain compatibility between solids and bars can be controlled to simulate perfect bonding or attend a bond–slip model. One disadvantage of the embedded method is the difficulty in applying boundary conditions at rebars since their displacement field is a function of the bulk elements displacements.

The first embedded technique is probably the one presented by Phillips and Zienkiewicz [12] and was applied to the simulation of RC structures. Nonetheless, the formulation is limited to 2D analyses, and perfect bonding is assumed between concrete and steel bars. Another limitation of the method is that bars are restricted to be parallel to the local coordinate axes of crossed elements. Later on, Elwi and Hrudey [13] provided a mathematical formulation to embed curved reinforcing bars into solids using strain compatibility and are among the first researchers to consider the prediction of reinforcement slippage based on extra degrees of freedom. Barzegar and Maddipudi [18,19] used an embedded approach based on the work of Elwi to simulate 3D RC structures and introduced a pre-processing tool to discretize straight reinforcements automatically. Hartl [14] simulated bond–slip in 3D RC structures using two approaches: one based on extra degrees of freedom and the other based on a secondary 1D truss–spring system where truss members represented the rebars, and the springs represented the bond with concrete. Jendele and Cervenka [15] simulated RC beams and slabs and introduced bond elements with an additional degree of freedom to account for slip; however, bars and bond elements are restricted to constant stresses. Lykidis and Spiliopoulos [20] analysed RC beam–column joints subjected to cyclic loads. To account for slip, they added extra degrees of freedom to elements containing rebars. In their approach, rebar strains were calculated from slip components and displacements interpolated from bulk elements. On the other hand, Dehestani and Mousavi [21] presented a modified steel bar model to account for slip. In that case, the authors modified the rebars strains by adding a portion corresponding to the slip.

To address the difficulties related to the bond–slip simulation in 3D and allow the application of boundary conditions to reinforcements in the embedded method, Durand [16] presented an approach using discrete rods and embedded link elements positioned at the rod nodes. In his method, the links are represented by springs whose elongation represents the slip. Also, boundary conditions can be directly applied to the rod nodes since they are modelled using discrete elements. This feature is very convenient in the simulation of pullout tests and prestressed concrete structures. Another approach using embedded link elements was presented by Zhou et al. [17], where 2D interface elements were used to simulate soil nailed structures. However, this approach cannot be naturally extended to three dimensions because the 2D interface elements must be located at either the upper or lower side of rebars. More recently, a method to link 3D embedded beam elements to solid finite elements through frictional contact has been proposed by Ninić et al. [22] and applied to soil piles. All interaction conditions are defined in the integration points of the beam elements representing the piles which are projected to the respective control points at the solid elements. In their formulation, an arbitrary number of piles with arbitrary orientation within one element can be considered. A frictional point-to-point contact algorithm was applied to model frictional contact between the soil and the embedded beam element using a tangent plane approximation describing the kinematics of multi-directional slip. The tip resistance at the pile foot is also accounted for by enforcing appropriate impenetrability conditions. Another recent work is the one presented by Bitencourt et al. [23]. In this technique, each rebar is discretized using nodes equally spaced. Later, the bulk elements that enclose these nodes are connected to the rebar nodes by coupling spring elements. The authors of [23] also presented a bond–slip model to simulate different bond conditions on 3D RC specimens. Finally, it is worth mentioning that other studies considering embedded rebars or fibres, including experimental works, are available in the literature [24–30].

This paper presents a novel method to simulate reinforced structures in 2D and 3D without the need for mesh compatibility. The method considers a discrete representation of the reinforcements using rod elements connected to bulk ones via novel rod–solid interface (or joint) elements. The interface elements enforce a continuous frictional contact that can be modelled using an arbitrary constitutive model. In particular, we present a cyclic constitutive model for the interface. By varying the interface properties along the rebar, it is possible to simulate fully anchored zones and free sliding zones in reinforced solids. Due to the discrete representation of reinforcements, forces or displacements can be prescribed at the rod elements' nodes. In addition, we present a tip bearing element designed to control the tip relative displacement in the reinforcement direction. This element covers two conditions: fully restrained tips (anchored tips) and tips restrained to only avoid penetration into the surrounding bulk element (non-anchored tips). Fig. 2 presents a schematic illustration of the element types involved in the method, including the novel interface and tip bearing elements.

Table 1
Feature comparison between the proposed and other methods.

Feature	Discrete	Embedded	Proposed
Reinforcements can be placed through bulk elements	No	Yes	Yes
Remeshing is required to test new reinforcement configurations	Yes	No	No
Extra DOFs are added to the system	Usually many, due to discretization constraints	Only if slip DOFs are considered	Only those from rod elements
Boundary conditions can be applied directly to rod elements	Yes	No	Yes
Straightforward simulation of the rod–solid contact	Yes, by the use of interface elements	No, usually slip DOFs are considered	Yes, by the use of the proposed interface element
The tip bearing effect can be considered at the ends of reinforcements	Yes, by the use of spring elements with variable stiffness	Only if slip DOFs are considered	Yes, by the use of the proposed tip bearing element

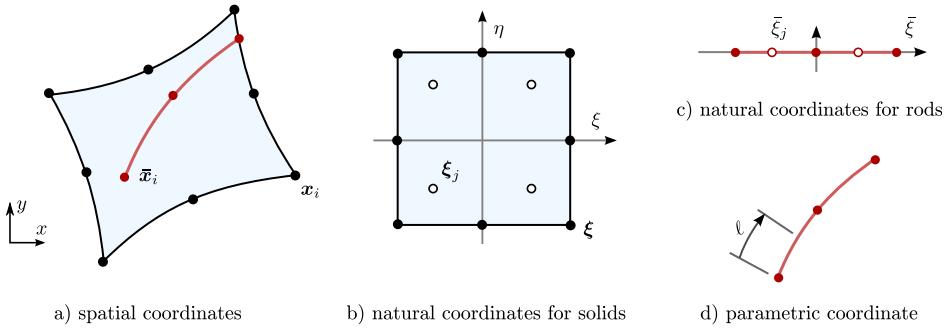


Fig. 3. Isoparametric mapping and notation to indicate coordinates in four different systems: (a) the spatial global coordinates system; (b) the reference system of the bulk element; (c) the reference system of the rod element; and (d) a local system aligned with the rod element with parametric coordinate ℓ. Filled circles symbolize nodes and empty circles integration points.

Table 1 shows a feature comparison between the proposed, the discrete, and the embedded methods. Essentially, the proposed method combines the positive features from both the discrete and embedded methods.

The paper is organized as follows. Section 2 presents the discretization and finite element formulation of curved rod elements that represent the reinforcements. Section 3 introduces the continuous interface element (rod–solid interface element) used to link a bulk element with a trespassing rod element. This section also introduces a cyclic bond–slip model. Section 4 introduces a tip bearing element used to restrain relative displacements at reinforcement tips. The finite element formulation for bulk elements is not covered in this paper since it is standard. The performance of the proposed method is demonstrated through a comprehensive set of numerical simulations of reinforced solids and structures in Section 5. The conclusions are given in Section 6, and the appendices present further details on the formulation used in the paper.

2. Curved rod element

This section deals with the discretization and finite element formulation of rod elements in the context of noncompatible reinforcements. Although the formulation of rod elements is classical, it is covered here for completeness and because it shares some aspects with the novel rod–solid interface element.

2.1. Reinforcement discretization

Prior to the finite element analyses, a pre-processing step is required to discretize reinforcement paths according to their intersections with the 2D or 3D bulk elements from the background mesh. A reinforcement path may be defined by a polyline, a polynomial or any other mathematical curve. The latter is important, for example, to consider curved reinforcements. Finding the intersection points, especially in meshes with nonlinear elements and curved reinforcement paths, is challenging. An efficient algorithm to accomplish this task was presented by Durand et al. [31] and is adopted in this paper.

Once the intersection points are found, the path can be subdivided into straight or curved rod elements. To illustrate the discretization process, Fig. 4(a) shows a background mesh and an arbitrary reinforcement path that is discretized using 2 and 3-node rod elements (Figs. 4(b) and 4(c)). Note that quadratic rod elements display a better representation of the original path. Finally, Fig. 4(d) shows a discretization using a background mesh composed of quadratic quadrilateral elements where crossed bulk elements are highlighted. Note that in this case, the bulk element edges are also curved. An important feature of the reinforcement discretization process is that the rod elements share the nodes located at the intersection points. This attribute guarantees that the displacement and axial stress fields are continuous between rod elements. Once the rod elements are defined, each rod–solid interface element is defined by taking the nodes from the corresponding rod and bulk elements.

2.2. Local coordinate system aligned with the rod

A local coordinate system aligned with the rod axis is defined to derive the rod element equations. Fig. 5 shows an orthonormal Cartesian system with basis $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$ established from the local Jacobian vector. In the following, a bar is used over most symbols related to the rod space. Before computing the rod's Jacobian, a conversion from the convected coordinates to the spatial coordinate system is applied. Using the isoparametric mapping, the position vector is given by

$$\bar{\boldsymbol{x}}(\ell) \approx \sum_m \bar{N}_m(\bar{\xi}(\ell))\, \bar{\boldsymbol{x}}_m \tag{1}$$

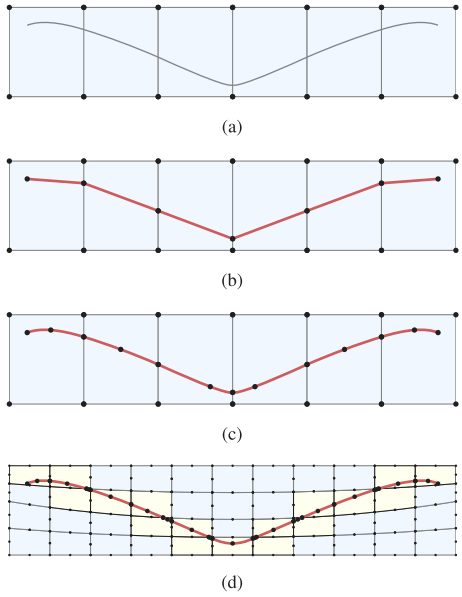


Fig. 4. Reinforcement discretization: (a) background mesh and a reinforcement path; (b) discretization using 2-node rod elements; (c) discretization using 3-node rod elements; (d) discretization using quadratic elements and a refined background mesh.

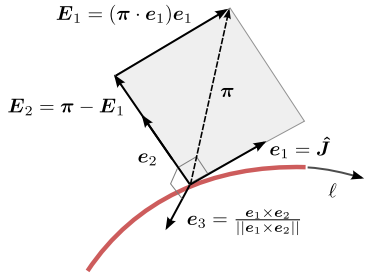


Fig. 5. Local orthonormal coordinate system aligned with the curve representing a rod element.

and the local Jacobian of the mapping yields

$$\boldsymbol{J}(\ell) = \frac{\mathrm{d}\bar{\boldsymbol{x}}}{\mathrm{d}\bar{\xi}} \approx \sum_m \frac{\mathrm{d}\bar{N}_m}{\mathrm{d}\bar{\xi}}\,\bar{\boldsymbol{x}}_m, \tag{2}$$

where ℓ is the longitudinal coordinate, ξ is the natural coordinate, and $\bar{N}$ is an interpolation function associated with the rod. Hence, the infinitesimal length $\mathrm{d}\ell$ along the curve can be calculated as follows

$$\mathrm{d}\ell \approx \|\mathrm{d}\bar{\boldsymbol{x}}\| = \|\boldsymbol{J}\|\,\mathrm{d}\bar{\xi} = J\,\mathrm{d}\bar{\xi} \tag{3}$$

Note that the basis $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$ varies along the parametric curve as well as the Jacobian. The base vector $\boldsymbol{e}_1$ is obtained as the unit Jacobian vector. The two remaining base vectors $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$ are found as follows (see Fig. 5). First, an auxiliary vector π, non-parallel to $\boldsymbol{J}$, is arbitrarily selected. The projection of π onto the direction of $\boldsymbol{e}_1$ gives $\boldsymbol{E}_1 = (\pi \cdot \boldsymbol{e}_1)\boldsymbol{e}_1$. Second, $\boldsymbol{E}_2$ is found by the parallelogram rule as $\boldsymbol{E}_2 = \pi - \boldsymbol{E}_1$. Next, $\boldsymbol{e}_2$ is obtained as the unit vector of $\boldsymbol{E}_2$. Finally, $\boldsymbol{e}_3$ is found as the unit vector from the cross product $\boldsymbol{e}_1 \times \boldsymbol{e}_2$. Note that $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$ are auxiliary vectors parallel to $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. The basis vectors can be arranged to build a rotation matrix $\boldsymbol{R}$ to express global displacements in the local system aligned with the rod. Thus,

$$\boldsymbol{R} = \begin{bmatrix} (\boldsymbol{e}_1)_1 & (\boldsymbol{e}_1)_2 & (\boldsymbol{e}_1)_3 \\ (\boldsymbol{e}_2)_1 & (\boldsymbol{e}_2)_2 & (\boldsymbol{e}_2)_3 \\ (\boldsymbol{e}_3)_1 & (\boldsymbol{e}_3)_2 & (\boldsymbol{e}_3)_3 \end{bmatrix}. \tag{4}$$

For instance, for an arbitrary displacements vector $\bar{\boldsymbol{u}}_i$ of a point and given in global coordinates, the corresponding displacements vector in the local system aligned with the rod is computed as $\bar{\boldsymbol{u}}_i^{e_1 e_2 e_3} = \boldsymbol{R}\,\bar{\boldsymbol{u}}_i$. This transformation will be used in the formulation of the rod–solid interface element, where relative displacements along the rod are transformed to the local system.

2.3. Axial strain

The strain at a point along the rod is defined as $\varepsilon = \frac{\mathrm{d}\bar{u}}{\mathrm{d}\ell}$, where $\bar{u}$ is the longitudinal displacement. In turn, $\bar{u}$ can be computed by interpolation of nodal displacements as

$$\bar{u} = \hat{\boldsymbol{J}} \cdot \left(\bar{\boldsymbol{N}}\bar{\boldsymbol{u}}\right), \tag{5}$$

where $\hat{\boldsymbol{J}}$ is the Jacobian unit vector, and $\bar{\boldsymbol{N}}$ is given by the matrix

$$\bar{\boldsymbol{N}} = \begin{bmatrix} \bar{N}_1\boldsymbol{I} & \bar{N}_2\boldsymbol{I} & \bar{N}_3\boldsymbol{I} & \cdots & \bar{N}_{\bar{n}}\boldsymbol{I} \end{bmatrix}, \tag{6}$$

where $\bar{n}$ is the rod's number of nodes and $\boldsymbol{I}$ is an identity matrix of order equal to the problem dimension. Eq. (5) can be conveniently expressed as $\bar{u} = \hat{\boldsymbol{J}}^T\left(\bar{\boldsymbol{N}}\bar{\boldsymbol{u}}\right)$; thus, the axial strain is

$$\varepsilon = \frac{\partial \bar{u}}{\partial \ell} = \hat{\boldsymbol{J}}^T \frac{\mathrm{d}\bar{\boldsymbol{N}}}{\mathrm{d}\ell}\bar{\boldsymbol{u}}. \tag{7}$$

Using the chain rule, this can be rewritten as

$$\varepsilon = \hat{\boldsymbol{J}}^T \frac{\mathrm{d}\bar{\boldsymbol{N}}}{\mathrm{d}\bar{\xi}} \frac{\mathrm{d}\bar{\xi}}{\mathrm{d}\ell}\bar{\boldsymbol{u}}, \tag{8}$$

where $\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}\ell}$ equals the inverse of the Jacobian norm (see Eq. (3)). Therefore, Eq. (8) is simplified to

$$\varepsilon = \frac{\boldsymbol{J}^T}{J^2} \frac{\mathrm{d}\bar{\boldsymbol{N}}}{\mathrm{d}\bar{\xi}}\bar{\boldsymbol{u}}. \tag{9}$$

From Eq. (9), we can conveniently extract the row matrix

$$\bar{\boldsymbol{B}} = \frac{\boldsymbol{J}^T}{J^2} \frac{\mathrm{d}\bar{\boldsymbol{N}}}{\mathrm{d}\bar{\xi}} \tag{10}$$

that relates the axial strain to the rod nodal displacements.

2.4. Internal force vectors

To derive the internal forces of the rod element, the internal force vectors $\bar{\boldsymbol{f}}_m$, corresponding to each node m of the rod element, are expressed as

$$\bar{\boldsymbol{f}}_m = \bar{f}_m(\ell)\,\hat{\boldsymbol{J}}, \tag{11}$$

where $\bar{f}_m$ is the axial force at node m and is given by

$$\bar{f}_m = \int_\ell \frac{\mathrm{d}\bar{N}_m}{\mathrm{d}\ell}\sigma A_s\,\mathrm{d}\ell. \tag{12}$$

Therein, $\sigma = \sigma(\ell)$ is the local axial stress, and A_s is the cross-section area of the rod. Combining Eq. (11) and (12) and with the aid of matrix $\bar{\boldsymbol{N}}$, the internal forces vector for the whole element is then written as

$$\bar{\boldsymbol{f}} = \int_\ell \frac{\mathrm{d}\bar{\boldsymbol{N}}^T}{\mathrm{d}\ell}\hat{\boldsymbol{J}}\sigma A_s\,\mathrm{d}\ell. \tag{13}$$

Taking into consideration $\frac{\mathrm{d}\bar{\boldsymbol{N}}}{\mathrm{d}\ell} = \frac{\mathrm{d}\bar{\boldsymbol{N}}}{\mathrm{d}\bar{\xi}}\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}\ell}$, $\hat{\boldsymbol{J}} = \boldsymbol{J}/J$ and $\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}\ell} = 1/J$, the resulting expression for the internal vector is

$$\bar{\boldsymbol{f}} = \int_\ell \left(\frac{\boldsymbol{J}^T}{J^2}\frac{\mathrm{d}\bar{\boldsymbol{N}}}{\mathrm{d}\bar{\xi}}\right)^T \sigma A_s\,\mathrm{d}\ell\,; \tag{14}$$

which, by incorporating Eq. (10), is simplified to

$$\bar{\boldsymbol{f}} = \int_\ell \bar{\boldsymbol{B}}^T \sigma A_s\,\mathrm{d}\ell. \tag{15}$$

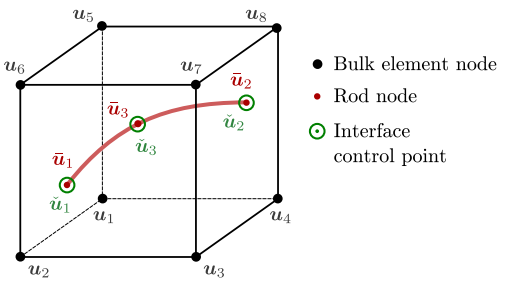


Fig. 6. Finite element representation of the interface element showing the rod nodes and corresponding control points located in the bulk element.

2.5. Stiffness matrix

The rod stiffness matrix is found by computing the derivative of $\bar{\boldsymbol{f}}$ (Eq. (15)) w.r.t. the nodal displacements $\bar{\boldsymbol{u}}$. To this end, the derivative of the axial stress w.r.t. the nodal displacements is obtained as

$$\frac{\mathrm{d}\bar{\sigma}}{\mathrm{d}\bar{\boldsymbol{u}}} = \frac{\mathrm{d}\bar{\sigma}}{\mathrm{d}\bar{\varepsilon}}\frac{\mathrm{d}\bar{\varepsilon}}{\mathrm{d}\bar{\boldsymbol{u}}} = E_s\bar{\boldsymbol{B}}\,, \tag{16}$$

where E_s is the reinforcement material's tangent stiffness that relates the axial stress to the axial strain. Thus, substituting Eq. (16) into $\frac{d\bar{f}}{d\bar{u}}$, the rod element stiffness matrix is given by

$$\bar{\boldsymbol{K}} = \int_{\ell} \bar{\boldsymbol{B}}^T\,\bar{\boldsymbol{B}}\,E_s A_s\,\mathrm{d}\ell. \tag{17}$$

3. Rod–solid interface element

A rod–solid interface element is introduced as an auxiliary entity to link a reinforcing rod element with the surrounding element. For illustration, Fig. 2 presents an analogy in which the surrounding solid material resembles a frictional casing enveloping the reinforcing bar. Fig. 6 presents the nodal displacements related to the interface element. For instance, the displacement vector of an interface element in 3D is given by

$$\begin{aligned}\tilde{\boldsymbol{u}} &= \{\boldsymbol{u}, \bar{\boldsymbol{u}}\}\\ &= \Big\{\underbrace{u_{1x}, u_{1y}, u_{1z}, u_{2x}, u_{2y}, u_{2z}, \ldots, u_{nx}, u_{ny}, u_{nz}}_{\boldsymbol{u}},\\ &\quad \underbrace{\bar{u}_{1x}, \bar{u}_{1y}, \bar{u}_{1z}, \bar{u}_{2x}, \bar{u}_{2y}, \bar{u}_{2z}, \cdots \bar{u}_{\bar{n}x}, \bar{u}_{\bar{n}y}, \bar{u}_{\bar{n}z}}_{\bar{\boldsymbol{u}}}\Big\}\end{aligned} \tag{18}$$

where n and $\bar{n}$ are the numbers of nodes from the solid and rod elements, respectively. Note that the total number of degrees of freedom is equal to the sum of those from solid and rod elements and that the rod–solid interface element does not add extra degrees of freedom to the system. The symbols $\check{\boldsymbol{u}}_1$, $\check{\boldsymbol{u}}_2, \ldots$ in Fig. 6 represent displacements of control points located inside the bulk element. They are used to compute the relative movement between the solid and the reinforcing rod.

3.1. Relative displacements

The relative displacements between the rod and the solid are required as input for the bond–slip model. To this end, virtual control points $\check{\boldsymbol{x}}_m$ located inside the bulk element are introduced (Fig. 6). They are positioned at the same spatial coordinates $\bar{\boldsymbol{x}}_m$ of the rod nodes. To keep the nomenclature short, in the following, the accent $\check{}$ is used over symbols related to the control points. A control point and corresponding rod node may develop different displacements when the rod slides due to axial and or solid deformations. The relative displacement vector, from the node i of the rod to the control point i, is defined as

$$\boldsymbol{w}_i = \bar{\boldsymbol{u}}_i - \check{\boldsymbol{u}}_i. \tag{19}$$

The displacements $\check{\boldsymbol{u}}_i$ of the control point i are computed by interpolation of nodal displacements from the bulk element as

$$\check{\boldsymbol{u}}_i \approx \sum_m N_m\left(\xi(\bar{\boldsymbol{x}}_i)\right)\,\boldsymbol{u}_m = \sum_m N_{m,i}\,\boldsymbol{u}_m\,, \tag{20}$$

where $N_{m,i}$ represent the mth shape function evaluated at the solid local coordinate $\xi(\bar{\boldsymbol{x}}_i)$ of the control point i. Note that $\bar{\boldsymbol{x}}_i = \check{\boldsymbol{x}}_i$. The process of finding $\xi(\bar{\boldsymbol{x}}_i)$ involves an *inverse mapping* procedure (see e.g. [13,14,31,32]) in which the spatial coordinates $\boldsymbol{x}_i$ are mapped into the reference coordinates system of the solid. The inverse mapping procedure used in this study is presented in Appendix A. For small strains, the coefficients $N_{m,i}$ in Eq. (20) can be computed just once during the initialization routine. The vector of displacements $\check{\boldsymbol{u}}$ from all control points can be expressed in matrix notation as

$$\begin{Bmatrix}\check{\boldsymbol{u}}_1\\ \check{\boldsymbol{u}}_2\\ \check{\boldsymbol{u}}_3\\ \vdots\\ \check{\boldsymbol{u}}_n\end{Bmatrix} = \begin{bmatrix} N_{1,1}\boldsymbol{I} & N_{2,1}\boldsymbol{I} & N_{3,1}\boldsymbol{I} & \cdots & N_{n,1}\boldsymbol{I}\\ N_{1,2}\boldsymbol{I} & N_{2,2}\boldsymbol{I} & N_{3,2}\boldsymbol{I} & \cdots & N_{n,2}\boldsymbol{I}\\ N_{1,3}\boldsymbol{I} & N_{2,3}\boldsymbol{I} & N_{3,3}\boldsymbol{I} & \cdots & N_{n,3}\boldsymbol{I}\\ \vdots & \vdots & \vdots & \ddots & \vdots\\ N_{1,\bar{n}}\boldsymbol{I} & N_{2,\bar{n}}\boldsymbol{I} & N_{3,\bar{n}}\boldsymbol{I} & \cdots & N_{n,\bar{n}}\boldsymbol{I}\end{bmatrix}\begin{Bmatrix}\boldsymbol{u}_1\\ \boldsymbol{u}_2\\ \boldsymbol{u}_3\\ \vdots\\ \boldsymbol{u}_n\end{Bmatrix}, \tag{21}$$

or in short,

$$\check{\boldsymbol{u}} = \boldsymbol{M}\boldsymbol{u}. \tag{22}$$

Once relative displacements at control points (Eq. (19)) are available, the relative displacements at any point along the rod can be computed by interpolation as:

$$\boldsymbol{w}(s) \approx \sum_m \bar{N}_m(\bar{\xi})\,\boldsymbol{w}_m\,, \tag{23}$$

where $\bar{\xi}$ is the natural coordinate in the rod system (see Fig. 3). These relative displacements can be expressed in the system $\boldsymbol{e}_i$ aligned with the rod (see Fig. 5), thus

$$\bar{w}_i(\bar{\xi}) = \boldsymbol{w}(\bar{\xi})\cdot\boldsymbol{e}_i(\bar{\xi}). \tag{24}$$

Note that the first component, $\bar{w}_1$, represents the rod slip. Combining Eqs. (19), (23) and (24), the vector with the local relative displacements corresponding to the system aligned with the rod is expressed as

$$\bar{\boldsymbol{w}} = \boldsymbol{R}\begin{bmatrix}-\bar{\boldsymbol{N}} & \bar{\boldsymbol{N}}\end{bmatrix}\begin{Bmatrix}\check{\boldsymbol{u}}\\ \bar{\boldsymbol{u}}\end{Bmatrix}, \tag{25}$$

where $\boldsymbol{R} = \boldsymbol{R}(\bar{\xi})$ is the rotation matrix introduced in Section 2.2 and $\bar{\boldsymbol{N}} = \bar{\boldsymbol{N}}(\bar{\xi})$ is the matrix given in Eq. (6). The substitution of Eq. (22) into Eq. (25) leads to

$$\bar{\boldsymbol{w}} = \boldsymbol{R}\begin{bmatrix}-\bar{\boldsymbol{N}}\boldsymbol{M} & \bar{\boldsymbol{N}}\end{bmatrix}\begin{Bmatrix}\boldsymbol{u}\\ \bar{\boldsymbol{u}}\end{Bmatrix}. \tag{26}$$

Note that $\bar{\boldsymbol{w}}$ is expressed as a function of the interface element displacements ($\boldsymbol{u}$ and $\bar{\boldsymbol{u}}$); thus, the matrix

$$\tilde{\boldsymbol{B}} = \boldsymbol{R}\begin{bmatrix}-\bar{\boldsymbol{N}}\boldsymbol{M} & \bar{\boldsymbol{N}}\end{bmatrix}, \tag{27}$$

can be used to relate the solid and rod displacements to relative displacements at a point with coordinate $\bar{\xi}$ in the reference system. It is worth noting that $\boldsymbol{R}$ and $\bar{\boldsymbol{N}}$ are evaluated at $\bar{\xi}$, and $\boldsymbol{M}$ at the bulk element natural coordinates ξ that correspond to the location of control points.

3.2. Contact force

Rod–solid contact forces are frequently considered in the literature as concentrated frictional forces, e.g. [16,22]. In this work, we consider frictional forces that are continuously distributed along the rod element. The force vector per unit length applied by the rod to the

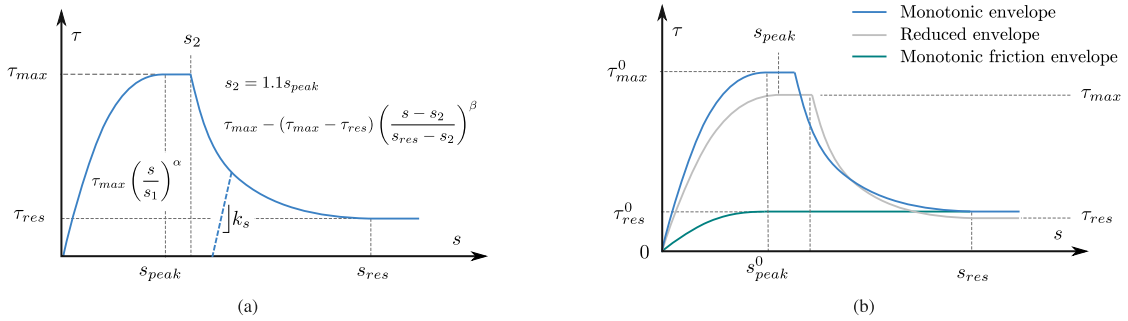


Fig. 7. Bond–slip model: (a) stress model; (b) envelopes for the total bond resistance along with the friction bond resistance envelope.

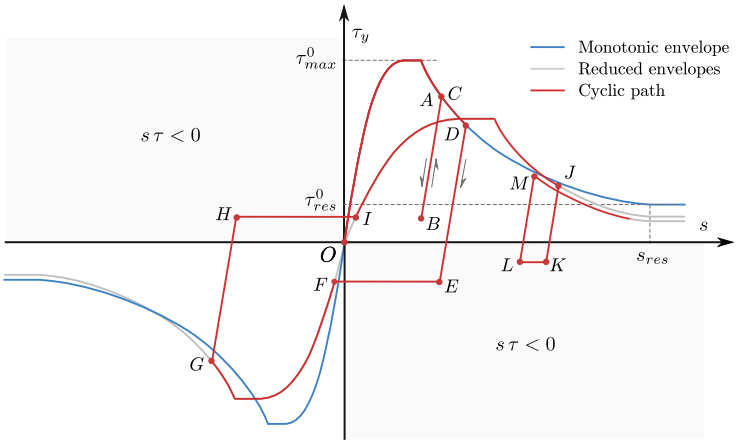


Fig. 8. Example of cyclic response using the proposed bond–slip model.

bulk element (i.e. the inner surface of the casing) is composed of three orthonormal components, thus

$$\dot{\boldsymbol{q}} = \dot{\tau}\, p\, \boldsymbol{e}_1 + \dot{q}_{n2}\, \boldsymbol{e}_2 + \dot{q}_{n3}\, \boldsymbol{e}_3, \tag{28}$$

where $\tau(\bar{w}_1)$ is the shear stress acting at the interface, p is the perimeter of the rod's cross-section, and $q_{n1}(\bar{w}_2)$ and $q_{n2}(\bar{w}_3)$ are distributed contact forces acting orthogonally to the rod. Assuming that each force component varies according to the corresponding relative displacement, we can write

$$\dot{\boldsymbol{q}} = \dot{\bar{w}}_1 \frac{d\tau}{d\bar{w}_1}\, p\, \boldsymbol{e}_1 + \dot{\bar{w}}_2\, k_n\, p\, \boldsymbol{e}_2 + \dot{\bar{w}}_3\, k_n\, p\, \boldsymbol{e}_3, \tag{29}$$

where $\frac{d\tau}{d\bar{w}_1}$ is a longitudinal tangent stiffness for the interface and k_n a penalty parameter for directions $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$.

Integrating $\boldsymbol{q}$ along the rod length, the forces at control points are found as

$$\check{\boldsymbol{f}} = \int_\ell \bar{\boldsymbol{N}}^T \boldsymbol{q}\, \mathrm{d}\ell. \tag{30}$$

Similarly, the forces applied by the casing to the surface of the rod are given by

$$\bar{\boldsymbol{f}} = -\check{\boldsymbol{f}}. \tag{31}$$

In turn, the nodal forces on the bulk element associated with the contact forces $\check{\boldsymbol{f}}$ are obtained from the bulk element's shape functions evaluated at the control points according to

$$\boldsymbol{f} = \boldsymbol{N}^T \check{\boldsymbol{f}}. \tag{32}$$

3.3. Consistent tangent matrix

During the solution of the nonlinear global system of equations with Newton's method, the derivative of the expression given by the difference between internal and external forces is required. Therefore, the derivative of the internal forces given in Eqs. (31) and (32) lead to the consistent tangent matrix

$$\tilde{\boldsymbol{K}} = \frac{\mathrm{d}\tilde{\boldsymbol{f}}}{\mathrm{d}\tilde{\boldsymbol{u}}} = \begin{bmatrix} \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}\boldsymbol{u}} & \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}\bar{\boldsymbol{u}}} \\ \frac{\mathrm{d}\bar{\boldsymbol{f}}}{\mathrm{d}\boldsymbol{u}} & \frac{\mathrm{d}\bar{\boldsymbol{f}}}{\mathrm{d}\bar{\boldsymbol{u}}} \end{bmatrix} \tag{33}$$

of the rod–solid interface element. The terms $\boldsymbol{f}$ and $\bar{\boldsymbol{f}}$ depend on $\check{\boldsymbol{f}}$, which in turn depends on $\boldsymbol{q}$. To compute the terms in Eq. (33), the derivative of $\boldsymbol{q}$ with respect to $\bar{\boldsymbol{w}}$ is required. This is given by

$$\frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}\bar{\boldsymbol{w}}} = p\boldsymbol{R}^T\tilde{\boldsymbol{D}}, \tag{34}$$

where $\tilde{\boldsymbol{D}}$ is the interface constitutive matrix and is expanded as

$$\tilde{\boldsymbol{D}} = \begin{bmatrix} \frac{d\tau}{d\bar{w}_1} & 0 & 0 \\ 0 & k_n & 0 \\ 0 & 0 & k_n \end{bmatrix}. \tag{35}$$

Therein, k_n is a penalty parameter chosen to provide negligible displacements in the directions $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$. For instance, we may consider $k_n = 10^4 \frac{d\tau}{d\bar{w}_1}$. For conventional RC structures, appropriate values are around 10^3 to 10^6 GPa. Considering the use of double-precision floating-point numbers, too large values may provide ill-conditioned systems of equations. In turn, $\frac{d\tau}{d\bar{w}_1}$ is obtained from experimental pullout curves. For perfect bonding conditions, this term can also be used as a penalty parameter and values in the same range as k_n can be used.

Using the chain rule, the derivatives of the internal forces related to the solid w.r.t. the solid and rod displacements are

$$\begin{aligned} \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}\boldsymbol{u}} &= \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}\check{\boldsymbol{f}}}\frac{\mathrm{d}\check{\boldsymbol{f}}}{\mathrm{d}\boldsymbol{q}}\frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}\bar{\boldsymbol{w}}}\frac{\mathrm{d}\bar{\boldsymbol{w}}}{\mathrm{d}\boldsymbol{u}} = \int_\ell \boldsymbol{N}^T\bar{\boldsymbol{N}}^T\boldsymbol{R}^T\tilde{\boldsymbol{D}}\ \left(\boldsymbol{R}\bar{\boldsymbol{N}}\boldsymbol{N}\right) p\, \mathrm{d}\ell \quad \text{and} \\ \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}\bar{\boldsymbol{u}}} &= \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}\check{\boldsymbol{f}}}\frac{\mathrm{d}\check{\boldsymbol{f}}}{\mathrm{d}\boldsymbol{q}}\frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}\bar{\boldsymbol{w}}}\frac{\mathrm{d}\bar{\boldsymbol{w}}}{\mathrm{d}\bar{\boldsymbol{u}}} = \int_\ell \boldsymbol{N}^T\bar{\boldsymbol{N}}^T\boldsymbol{R}^T\tilde{\boldsymbol{D}}\ \left(-\boldsymbol{R}\bar{\boldsymbol{N}}\right) p\, \mathrm{d}\ell, \end{aligned} \tag{36}$$

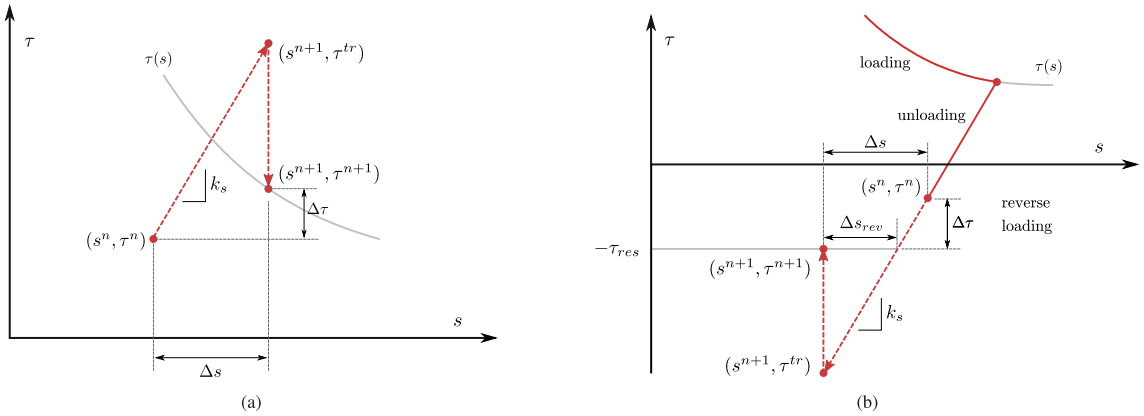


Fig. 9. Nonlinear stress update for the bond–slip model: (a) for $s^{n+1}\,\tau^{tr} \geq 0$; (b) for $s^{n+1}\,\tau^{tr} < 0$.

and the derivatives of the internal forces related to the rod w.r.t. the solid and rod displacements are

$$\begin{aligned}\frac{\mathrm{d}\bar{\boldsymbol{f}}}{\mathrm{d}\boldsymbol{u}} &= \frac{\mathrm{d}\bar{\boldsymbol{f}}}{\mathrm{d}\check{\boldsymbol{f}}}\frac{\mathrm{d}\check{\boldsymbol{f}}}{\mathrm{d}\boldsymbol{q}}\frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}\bar{\boldsymbol{w}}}\frac{\mathrm{d}\bar{\boldsymbol{w}}}{\mathrm{d}\boldsymbol{u}} = \int_{\ell} \bar{\boldsymbol{N}}^T \boldsymbol{R}^T \tilde{\boldsymbol{D}} \left(\boldsymbol{R}\bar{\boldsymbol{N}}\boldsymbol{N}\right) p\,\mathrm{d}\ell \quad \text{and} \\ \frac{\mathrm{d}\bar{\boldsymbol{f}}}{\mathrm{d}\bar{\boldsymbol{u}}} &= \frac{\mathrm{d}\bar{\boldsymbol{f}}}{\mathrm{d}\check{\boldsymbol{f}}}\frac{\mathrm{d}\check{\boldsymbol{f}}}{\mathrm{d}\boldsymbol{q}}\frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}\bar{\boldsymbol{w}}}\frac{\mathrm{d}\bar{\boldsymbol{w}}}{\mathrm{d}\bar{\boldsymbol{u}}} = \int_{\ell} \bar{\boldsymbol{N}}^T \boldsymbol{R}^T \tilde{\boldsymbol{D}} \left(-\boldsymbol{R}\bar{\boldsymbol{N}}\right) p\,\mathrm{d}\ell.\end{aligned} \tag{37}$$

Substituting the above derivatives into Eq. (33), and after some rearrangements, the stiffness matrix of the rod–solid interface element is finally expressed as

$$\tilde{\boldsymbol{K}} = \int_{\ell} \tilde{\boldsymbol{B}}^T \tilde{\boldsymbol{D}} \tilde{\boldsymbol{B}} p\,\mathrm{d}\ell. \tag{38}$$

It is worth mentioning that the rod–solid interface stiffness matrix needs to be integrated numerically along the rod length. However, the number of integration points of the rod–solid interface element can differ from the number of integration points used in the rod element due to the higher degree of the integrand. In some cases, the rod–solid interface stiffness matrix may show rank deficiency if insufficient integration points are used, leading to spurious displacement modes. In this regard, we recommend using two integration points for linear interpolation and three integration points for quadratic interpolation. Note also that the proposed interface element is different from spring elements. The contact equations are derived from the equilibrium along the rod and not from single points, as in the case of spring elements. This feature provides more accuracy since it can simulate contact stress distributions along the rod. Algorithm 1 concisely presents the computation of the rod–solid stiffness matrix in the context of numerical integration.

Algorithm 1: Computation of the rod–solid interface element stiffness matrix.

Data: current interface element data
Result: $\tilde{\boldsymbol{K}}$
$\tilde{\boldsymbol{K}} = \boldsymbol{0}$ *! stiffness matrix initialization*
for each interface integration point ($\bar{\xi}_i$) along the rod **do**
  $\boldsymbol{J} = \boldsymbol{J}(\bar{\xi}_i)$ *! compute according to Eq. (2)*
  $\boldsymbol{R} = \boldsymbol{R}(\bar{\xi}_i)$ *! compute according to Eq. (4)*
  $\bar{\boldsymbol{N}} = \bar{\boldsymbol{N}}(\bar{\xi}_i)$ *! compute according to Eq. (6)*
  $\xi = \xi(\bar{\boldsymbol{x}}_i(\bar{\xi}_i))$ *! control points coords. in the solid reference*
  $\boldsymbol{M} = \boldsymbol{M}(\xi)$ *! compute according to Eq. (21)*
  $\tilde{\boldsymbol{B}} = \boldsymbol{R}\begin{bmatrix}-\bar{\boldsymbol{N}}\boldsymbol{M} & \bar{\boldsymbol{N}}\end{bmatrix}$
  $\tilde{\boldsymbol{D}} = \tilde{\boldsymbol{D}}(\bar{\xi}_i)$ *! compute according to Eq. (35)*
  $w = w(\bar{\xi}_i)$ *! integration point weight*
  $\tilde{\boldsymbol{K}} := \tilde{\boldsymbol{K}} + \tilde{\boldsymbol{B}}^T \tilde{\boldsymbol{D}} \tilde{\boldsymbol{B}}\, p\, w\, \|\boldsymbol{J}\|$

3.4. Cyclic bond–slip constitutive model

The constitutive model for the bond–slip behaviour relates the longitudinal shear stress τ with the relative displacement $\bar{w}_1$ (slip). In this section, we adopt $s = \bar{w}_1$ for simplicity and to agree with the nomenclature of existing literature. Different models can be used in conjunction with the proposed approach. The model presented here (Fig. 7(a)) is based on the monotonic envelope of the FIB model [33] and the cyclic model presented by Murcia-Delso et al. [34]. Note that the monotonic response proposed by the FIB model is intended to represent concrete–steel bond–slip behaviour. However, similar responses may apply to other materials. In the proposed model, the shear strength is given by a piecewise function defined as

$$\begin{aligned}&\tau(s) = \tau_{max}\left(\frac{s}{s_{peak}}\right)^{\alpha} && \text{for} \quad s \leq s_{peak}, \\ &\tau(s) = \tau_{max} && \text{for} \quad s_{peak} < s \leq 1.1 s_{peak}, \\ &\tau(s) = \tau_{max} - \left(\tau_{max} - \tau_{res}\right) \\ &\qquad \times \left(\frac{s - 1.1 s_{peak}}{s_{res} - 1.1 s_{peak}}\right)^{\beta} && \text{for} \quad 1.1 s_{peak} < s \leq s_{res} \quad \text{and} \\ &\tau(s) = \tau_{res} && \text{for} \quad s_{res} < s.\end{aligned} \tag{39}$$

In the equation above, τ_{max} is the maximum strength, τ_{res} is the residual strength, α and β are parameters that define the curvature of the ascending and descending branches, respectively, and s_{peak} and s_{res} are characteristic slip values according to Fig. 7(a). In elastic conditions, the increments of slip and shear stress are related by the elastic stiffness k_s. Fig. 7(b) shows the monotonic envelope where τ^0_{max}, s^0_{peak} and s^0_{peak} represent the initial curve parameters. Note that k_s is always less than the initial tangent stiffness. In cyclic loadings, these parameters are modified to produce reduced envelopes to account for the bond damage. The reduced envelopes are given by the same function in Eq. (39) but are computed using modified values for τ_{max}, τ_{res} and s_{peak} according to the amount of slip reversal s_{rev} and the slip amplitude s_{amp} (maximum positive slip s^+_{max} plus maximum negative slip s^-_{max}). Fig. 7(b) also displays the monotonic friction envelope that represents the evolution of the residual strength. The curved section of the friction envelope follows a similar curve as the ascending branch of the total bond resistance, thus

$$\tau_{res} = \tau^0_{res}\left(\frac{s}{s^0_{peak}}\right)^{\alpha} \tag{40}$$

Fig. 8 presents an example path to illustrate the model response under cyclic loadings. The loading path starts at point O and follows the monotonic envelope up to point A. Next, the trajectory ABC represents an elastic unloading–reloading loop. After point C, the path continues over the monotonic curve up to point D when unloading and reverse

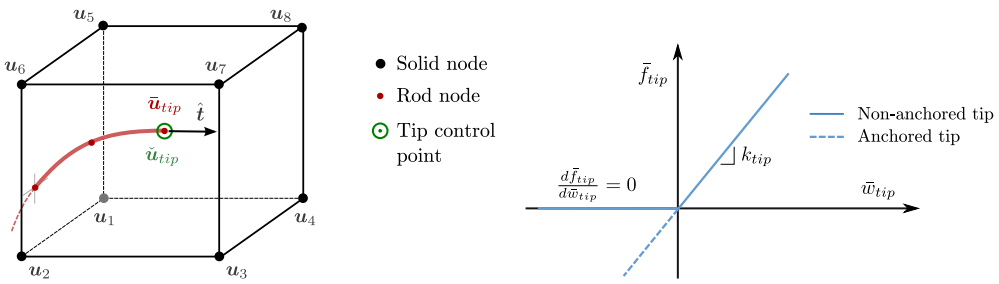


Fig. 10. Tip bearing element: (a) finite element representation showing the tip node and corresponding control point located in the bulk element; (b) bearing stiffness at the tip.

loading occurs until reaching a stress value equal to $-\tau_{res}$ at point E. The path continues under slip reversal constrained by the residual strength until the intersection with the reduced envelope at point F. Then, the path continues until point G when a subsequent loading leads to the updated residual stress $+\tau_{res}$ at point H. The path continues until the intersection with a second reduced envelope at point I and follows this curve until point J. The trajectory $JKLM$ represents another loop and includes reverse slip. After point M, the loading path follows another reduced envelope. The behaviour of the presented cyclic model is compatible with experimental observations (see e.g. [34,35]).

To take into account the history of cycles in the definition of reduced envelopes, a normalized slip variable that considers the amount of slip reversal and the slip amplitude is proposed as

$$\hat{s} = \frac{s_{amp} s_{rev}}{(s^0_{peak})^2}, \tag{41}$$

where s^0_{peak} is the slip value corresponding to the peak of the initial envelope. Then, under cyclic loadings, the envelope parameters τ_{max}, τ_{res} and s_{peak} in Eq. (39) are recomputed according to the following proposed expressions

$$\tau_{max} = \tau^0_{max}\, e^{-0.015\,\hat{s}}, \tag{42}$$

$$s_{peak} = s^0_{peak}\left(1 + 0.33\sqrt{\hat{s}}\right) \tag{43}$$

and

$$\tau_{res} = \begin{cases} \tau^0_{res}\left(\frac{s}{s^0_{peak}}\right)^{\alpha} & \text{for} \quad s_{acc} < s_{peak} \\ \tau^0_{res}\, e^{-0.0027\,\hat{s}} & \text{for} \quad s_{acc} \geq s_{peak} \end{cases} \tag{44}$$

where s_{acc} represents the absolute accumulated slip.

Finally, the stress integration step in a finite element analysis proceeds as follows, First, for a given increment Δs at step n, the elastic trial stress is given by

$$\tau^{tr} = \tau^n + k_s\, \Delta s. \tag{45}$$

Then, considering $s^{n+1} = s^n + \Delta s$, a nonlinear stress prediction (absolute value) is computed according to the quadrant where the point (s^{n+1}, τ_{tr}) is located, thus

$$\tau^{n+1}_{nl} = \begin{cases} \tau_{res} & \text{for} \quad s^{n+1}\,\tau^{tr} < 0 \\ \max(\tau_{res}, \tau(s^{n+1})) & \text{for} \quad s^{n+1}\,\tau^{tr} \geq 0 \end{cases} \tag{46}$$

where $\tau(s^{n+1})$ is evaluated according to Eq. (39). Next, if $|\tau_{tr}| < \tau^{n+1}_{nl}$, the increment is elastic and $\tau^{n+1} = \tau_{tr}$. Otherwise, the increment is nonlinear and $\tau^{n+1} = \tau^{n+1}_{nl} \operatorname{sgn} \tau^{tr}$. Fig. 9 shows two cases of nonlinear stress update according to the sign of $s^{n+1}\,\tau^{tr}$. In the case of $s^{n+1}\,\tau^{tr} < 0$, the value for the reverse slip s_{rev} is updated using the increment

$$\Delta s_{rev} = \frac{|\tau^{tr}| - \tau^{n+1}_{nl}}{k_s}. \tag{47}$$

The stress update process for the bond–slip model is summarized in Algorithm 2.

Algorithm 2: Stress update for the bond–slip model.

Data: τ^n, s^n, Δs, s_{acc}, s_{rev}, s^+_{max}, s^-_{max}
Result: τ^{n+1}, $\Delta\tau$
$s^{n+1} = s^n + \Delta s$ *! updated slip*
$s_{acc} = s_{acc} + |\Delta s|$ *! accumulated absolute slip*
$s^+_{max} = \max(s^+_{max}, s^n)$ *! maximum positive slip*
$s^-_{max} = -\min(-s^-_{max}, s^n)$ *! maximum negative slip*
$s_{amp} = s^+_{max} + s^-_{max}$ *! slip amplitude*
! Modified curve parameters
$\hat{s} = \frac{s_{amp} s_{rev}}{\left(s^0_{peak}\right)^2}$
$\tau_{max} = \tau^0_{max}\, e^{-0.15\hat{s}}$
$s_{peak} = s^0_{peak}\left(1 + 0.33\hat{s}^{0.5}\right)$
if $s_{acc} < s_{peak}$ **then**
  $\tau_{res} = \tau^0_{res}\left(\frac{s_{acc}}{s_{peak}}\right)^{\alpha}$
else
  $\tau_{res} = \tau^0_{res}\, e^{-0.0027\hat{s}}$
! Nonlinear stress prediction
$\tau^{tr} = \tau^n + k_s\, \Delta s$ *! elastic trial*
if $s^{n+1}\,\tau^{tr} < 0$ **then**
  $\tau^{n+1}_{nl} = \tau_{res}$
else
  $\tau^{n+1}_{nl} = \max(\tau_{res}, \tau(s^{n+1}))$
! Stress updating
if $|\tau^{tr}| \leq \tau^{n+1}_{nl}$ **then** *! linear increment*
  $\tau^{n+1} = \tau^{tr}$
else *! nonlinear increment*
  if $s^{n+1}\,\tau^{tr} < 0$ **then**
    $\Delta s_{rev} = (|\tau_{tr}| - \tau^{n+1}_{nl})/k_s$
    $s_{rev} = s_{rev} + \Delta s_{rev}$
  $\tau^{n+1} = \operatorname{sgn}(\tau^{tr})\, \tau^{n+1}_{nl}$ *! updated stress*
$\Delta\tau = \tau^{n+1} - \tau^n$ *! stress increment*

4. Tip bearing element

The proposed tip bearing element is designed to limit the rod's tip relative displacement in the reinforcement direction. This element covers two conditions: (1) fully restrained tips (anchored tips) and (2) tips restrained to avoid penetration into the reinforced medium (non-anchored tips). The relative displacement at the tip is computed from the displacements of the tip node and a control point located inside the bulk element. Fig. 10(a) shows a finite element representation of the tip node and the corresponding control point. Regarding the degrees of freedom of a tip bearing element, the displacements vector in 3D is

Table 2
Example 1 — Patch test. Material parameters.

Solids	$E = 1$	$\nu = 0.25$	
Bars	$E_s = 10$	$A_s = 0.01$	
Interfaces	$A = 0.01$	$k_s = 100$	$k_n = 100$

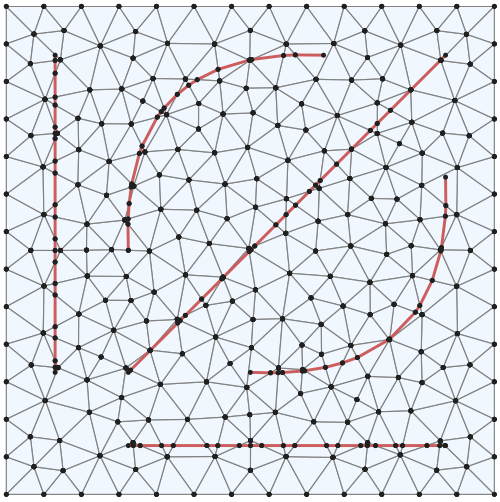


Fig. 11. Example — Patch test. Finite element mesh showing the reinforcing bars and intersection points.

given by

$$\tilde{\boldsymbol{u}}_{tip} = \left\{\boldsymbol{u}, \bar{\boldsymbol{u}}_{tip}\right\} = \left\{ \underbrace{u_{1x}, u_{1y}, u_{1z}, u_{2x}, u_{2y}, u_{2z}, \ldots, u_{nx}, u_{ny}, u_{nz}}_{\boldsymbol{u}}, \underbrace{\bar{u}_{tip\,x}, \bar{u}_{tip\,y}, \bar{u}_{tip\,z}}_{\bar{\boldsymbol{u}}_{tip}} \right\} \tag{48}$$

where $\boldsymbol{u}$ is the vector with nodal displacements from the bulk element and $\bar{\boldsymbol{u}}_{tip}$ is a vector with the tip node displacements.

The tip bearing elements can be used conveniently at one or the two reinforcement endpoints. For instance, the simulation of a rebar immersed in a compressed material requires two non-anchored tip bearing elements. In turn, in a rebar with a fixed end (e.g. welded or clamped), an anchored tip bearing element is required to simulate the tip interaction. Conversely, the simulation of a pullout test does not require the use of any tip bearing elements. In cases where the reinforcement geometry is represented by a polyline (e.g. stirrups), no tip bearing elements are required at the corners since the nonaxial stiffness components from concurrent rod-interface elements at these points will prevent the penetration of rod elements.

4.1. Stiffness matrix

Analogous to Eq. (19), the relative displacement between the tip and the corresponding control point is computed as

$$\boldsymbol{w}_{tip} = \bar{\boldsymbol{u}}_{tip} - \check{\boldsymbol{u}}_{tip} \tag{49}$$

where $\bar{\boldsymbol{u}}_{tip}$ is a vector with the tip displacements and $\check{\boldsymbol{u}}_{tip}$ a vector with the displacements at the control point. $\check{\boldsymbol{u}}_{tip}$ can be computed by interpolation of the bulk element displacements as

$$\check{\boldsymbol{u}}_{tip} = \boldsymbol{M}_{tip}\boldsymbol{u}. \tag{50}$$

Therein, matrix $\boldsymbol{M}_{tip}$ is given by

$$\boldsymbol{M}_{tip} = \begin{bmatrix} N_{1,tip}\boldsymbol{I} & N_{2,tip}\boldsymbol{I} & N_{3,tip}\boldsymbol{I} & \cdots & N_{n,tip}\boldsymbol{I} \end{bmatrix}, \tag{51}$$

where $N_{m,tip}$ represents the mth shape function from the bulk element evaluated at the natural coordinates corresponding to the rod tip, $\xi(\boldsymbol{x}_{tip})$. Combining Eqs. (49) and (50), we can express the relative displacement at the tip in the direction aligned with the rod as

$$\bar{w}_{tip} = \hat{\boldsymbol{t}}^T \begin{bmatrix} -\boldsymbol{M}_{tip} & \boldsymbol{I} \end{bmatrix} \begin{Bmatrix} \boldsymbol{u} \\ \bar{\boldsymbol{u}}_{tip} \end{Bmatrix}, \tag{52}$$

where $\hat{\boldsymbol{t}}$ is a unit vector aligned with the rod at the tip location (see Fig. 10(a)). From Eq. (52) we obtain the matrix

$$\boldsymbol{B}_{tip} = \hat{\boldsymbol{t}}^T \begin{bmatrix} -\boldsymbol{M}_{tip} & \boldsymbol{I} \end{bmatrix} \tag{53}$$

that relates the nodal displacements from the tip bearing element with the relative displacement at the tip, thus $w_{tip} = \boldsymbol{B}_{tip}\tilde{\boldsymbol{u}}_{tip}$. Likewise, the vector of nodal forces and the force at the tip can be expressed as $\tilde{\boldsymbol{f}}_{tip} = \bar{f}_{tip}\boldsymbol{B}_{tip}^T$, where $\bar{f}_{tip}$ is the reaction force at the tip. Adopting a procedure similar to the rod–solid interface element, the stiffness matrix for the tip bearing element is found as

$$\tilde{\boldsymbol{K}}_{tip} = \frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}\boldsymbol{B}_{tip}^T\boldsymbol{B}_{tip}, \tag{54}$$

where $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}$ is the tangent stiffness that relates increments of force and relative displacement at the bearing tip. If the tip is anchored, the value of $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}$ is given by a penalty parameter k_{tip}. This parameter is conveniently chosen to provide a negligible tip displacement. For conventional RC structures, good results were obtained using k_{tip} between 10^6 to 10^9 kN/m. This range is compatible with the recommendation proposed by Bitencourt et al. [23] for coupling springs in perfect bonding conditions. For the case of non-anchored tips, a nonlinear force–displacement law is required.

4.2. Force–displacement model

In a non-anchored tip, the tip bearing element is activated only when the rod tends to penetrate the reinforced medium. To identify this condition, the auxiliary vector $\hat{\boldsymbol{t}}$ is considered to point outwards the rod (Fig. 10(a)). Considering that the rod Jacobian vector $\boldsymbol{J}$ provides the rod orientation, vector $\boldsymbol{t}$ can be computed as follows. If the bearing tip node is the same as the first rod node, then $\hat{\boldsymbol{t}} = -\hat{\boldsymbol{J}}$, otherwise $\hat{\boldsymbol{t}} = \hat{\boldsymbol{J}}$. In this case, $\hat{\boldsymbol{J}}$ is the Jacobian unit vector evaluated at the tip natural coordinates. Following this procedure, Eq. (52) provides a positive relative displacement if the tip moves in the direction of $\hat{\boldsymbol{t}}$. This feature is used to define the tip bearing element behaviour as shown in Fig. 10(b). Thus, if $\bar{w}_{tip} > 0$, $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}$ is assumed equal to a penalty parameter k_{tip} and the stiffness matrix is computed as Eq. (54). Otherwise, $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}} = 0$, leading to $\tilde{\boldsymbol{K}}_{tip} = \boldsymbol{0}$.

Finally, the stress integration step in a finite element analysis must be considered. Given the increment $\Delta\bar{w}$ at step n, the force increment at the tip is updated as follows. First, a trial value is computed $f_{tip}^{tr} = f_{tip}^{n} + k_{tip}\Delta_w$, where f_{tip}^{n} is the force at step n. Next, if $f_{tip}^{tr} > 0$, the force is updated as $f_{tip}^{n+1} = f_{tip}^{tr}$. Otherwise, it is updated as $f_{tip}^{n+1} = 0$.

5. Numerical examples

In this section, the performance of the rod–solid interface element is demonstrated through a comprehensive set of numerical simulations of reinforced solids and structures. First, a patch test is performed to verify that no spurious displacements are introduced at the rod or interface elements by applying a rigid-body motion to the reinforced medium. Next, two tests previously studied by other authors are analysed. To assess the method accuracy, numerical stresses from a reinforced beam and a truss located inside a soft medium are compared with the analytical solutions. The behaviour of the proposed constitutive model for the interface is studied through the simulation of a pullout experiment, a cube reinforced by an internal spiral and a cyclic pullout experiment. The convergence properties are assessed through a set of simulations

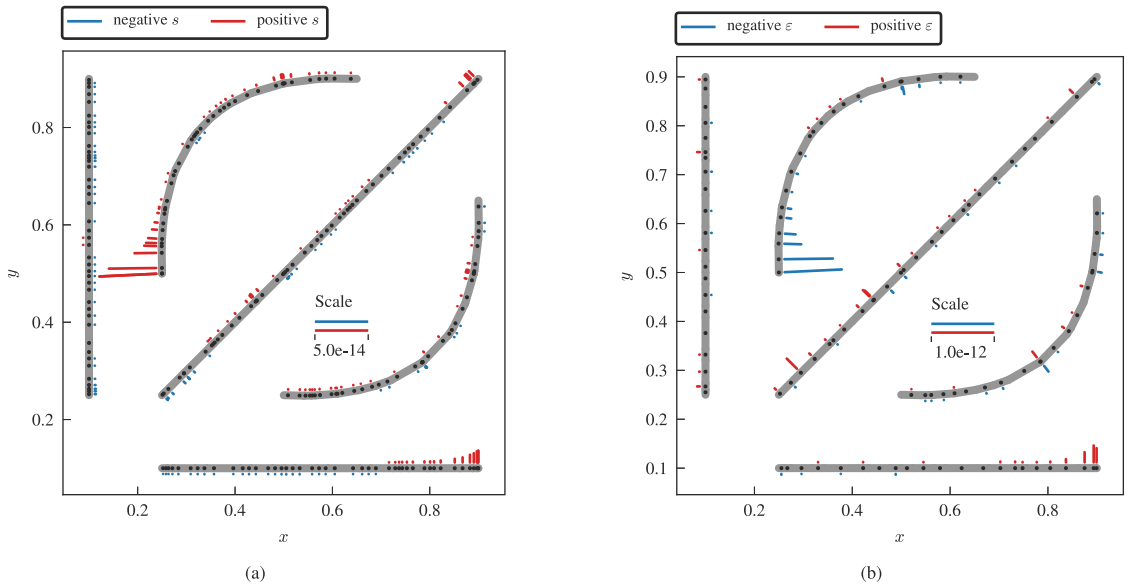


Fig. 12. Example — Rigid motion patch test: (a) slip values computed at the integration points of interface elements; (b) axial strains at the integration points of rod elements.

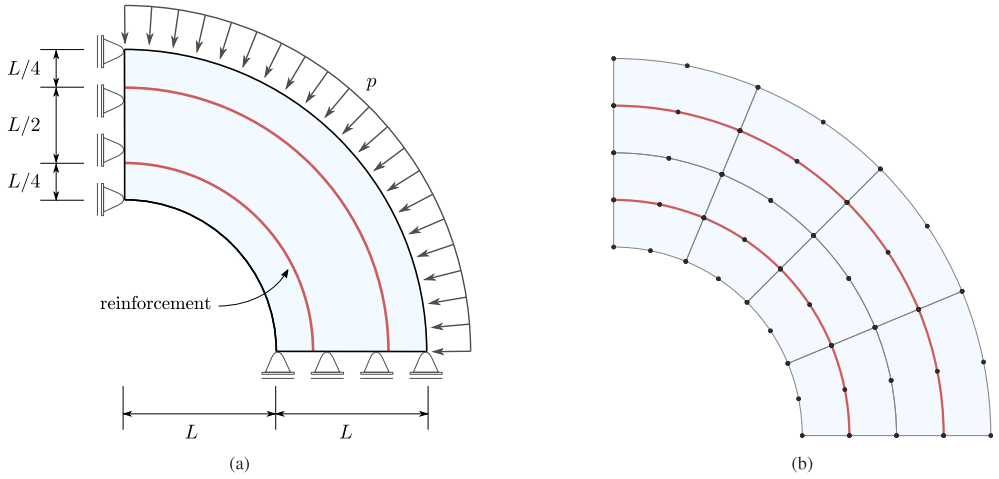


Fig. 13. Example — Quarter ring with two curved reinforcing layers: (a) test setup; (b) finite element mesh.

of a reinforced beam and a pullout test using different meshes. Finally, the proposed method is used in the simulation of a reinforced concrete corbel. A future publication is on the works with an extensive study of the new technique applied to the simulation of reinforced concrete structures.

Unstructured and non-structured meshes are used to represent the reinforced media. Non-structured meshes are generated using the Gmsh [36] and TetGen [37–39] mesh generators. The simulations are performed using the Amaru library (see [40] and https://github.com/NumSoftware/Amaru), which includes the implementation of the proposed interface element and constitutive model. The algorithm presented by Durand et al. [31] is used for the reinforcement discretization.

5.1. Example 1 — Patch test

To verify the validity of the proposed formulation, a patch test in a reinforced medium is conducted. The test aims to verify that no spurious displacements are introduced at the rod or interface elements by applying rigid-body motion at the reinforced medium. The domain is given by a reinforced square with sides equal to 1. Straight and curved bars with different orientations are used to check any possible effect. The finite element mesh is presented in Fig. 11 and is constituted by 3-node triangle elements and 2-node rod elements, all with one integration point. The rod–solid interface elements are modelled using linear interpolation and two integration points. The material properties used are presented in Table 2. All materials are considered linear elastic. The boundary conditions are specified to impose a constant displacement field in the reinforced medium; thus, all nodes from triangular elements are subjected to $u_x = 1$ and $u_y = 1$. No displacement boundary conditions are applied to the rod or interface elements.

Fig. 12 shows the relative longitudinal displacements and strains obtained at integration points of the interface and rod elements, respectively. The length of the perpendicular lines represents the relative magnitude of the quantity being plotted. The results show negligible values, in the order of 10^{-14} and 10^{-12}, for the relative displacements and axial strains, respectively. This fact demonstrates that the proposed formulation does not introduce significant displacements to rod or interface elements.

5.2. Example 2 — Quarter ring with two curved reinforcing layers

This example presents a quarter ring with two reinforcing layers subjected to a radial load, as shown in Fig. 13(a). This test was initially studied by Elwi and Hrudey [13] using the embedded method and, more recently, by Bitencourt et al. [23] using a method that involves coupling elements. Perfect bonding and plane-stress conditions are considered. To assist in the comparison of results, the mesh used in the original work is considered here as well (Fig. 13(b)). The mesh comprises 8-node quadrilateral elements with four integration points and 3-node rods with two integration points. Also, quadratic interface elements with three integration points are considered to link bulk to rod elements. The material parameters are linear elastic and are given by $E = 2.8{\times}10^3$ MPa, $E_s = 8E$ and $\nu = 0.25$. The length and width of the plate are taken as $L = 1$ m, as well as the thickness. The cross-sectional area for each bar is $A_s = 0.025\,\mathrm{m}^2$. The radial loading was considered equal to $p = 1$ kPa. To simulate perfect bonding, $k_s = k_n = 10^3$ GPa/m was considered.

Fig. 14 shows the axial stress values computed at the integration points (empty red circles) from both reinforcing layers. For the sake of comparison, the figure also shows the results obtained by other authors, along with the results using a refined mesh after a convergence test. At both the inner and outer layers, the coarse mesh results are in excellent agreement with those from the refined mesh. Also, they are in good agreement with the results presented by Elwi and Hrudey. Note that the results from Bitencourt et al. are in good agreement with those from the refined mesh; however, the stress values present an irregular pattern, especially at the inner layer.

5.3. Example 3 — Square plate subjected to uniaxial tensile load

In this test, a square plate with two reinforcing bars is analysed under plane-stress conditions. No bond–slip interaction is considered. This problem was previously investigated by Elwi and Hrudey [13] using the embedded method and later by El-Mezaini and Çitipitioğlu [8] using the discrete method with elements where mid-side nodes are adjusted to match the reinforcement nodes. According to Fig. 15(a), the plate is fixed along the left side, and a uniform load $p = 1$ kPa is applied on the right side. No load is applied to the reinforcing bars. The material parameters are the same as in Example 1. A relatively coarse mesh is employed (see Fig. 15(b)), which corresponds to the one used in the aforementioned analyses. Quadratic interpolation is used for bulk and rod elements, with 8 and 2 integration points, respectively. The interface elements are also assumed quadratic and have two integration points.

Fig. 16 shows the distribution of the axial stresses in the reinforcement bars evaluated at the integration points of the rod elements (red circles). For comparison, the figure also includes a curve obtained using a refined mesh (containing 40 × 40 elements) after performing a convergence analysis and the results from the aforementioned authors. The results from the coarse mesh show an excellent agreement with the results from the refined one. In comparison, the discrete method with movable mid-side nodes [8] provide similar results as the proposed method; however, some discrepancies are observed near the fixed boundary. Those discrepancies may be related to the use of a coarse mesh. On the other hand, the stress distribution obtained using the embedded method with no bond–slip [13] is inaccurate and erroneously constant. This is because in that method, the effect of reinforcing bars is considered by simply increasing the stiffness of crossed bulk elements, losing the local interaction between rebars and the surrounding material.

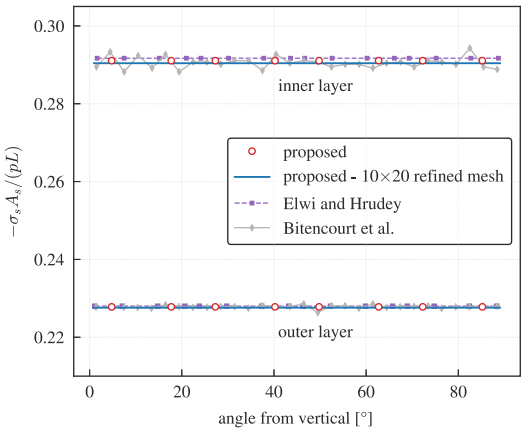


Fig. 14. Example 3 — Quarter ring with two curved reinforcing layers. Normalized axial stresses at the integration points of the rod elements compared with the results using a refined mesh and the results from Elwi and Hrudey [13] and Bitencourt et al. [23].

Table 3
Example 4 — Truss structure inside in a very soft material. Material parameters.

Solids	$E = 1$ kPa	$\nu = 0.25$	
Bars	$E_s = 210$ GPa	$A_s = 0.001$ m^2	
Interface	$A = 0.001$ m^2	$k_s = 10^3$ GPa/m	$k_n = 10^3$ GPa/m

5.4. Example 4 — Truss structure inside a very soft material

To further assess the predictive capabilities of the proposed method, a 2D frame structure inside a very soft material (low stiffness) is simulated using a (challenging) non-structured 3D mesh. A similar test was proposed by Hartl [14]; however, using very few elements. This test aims to reproduce the axial stresses in a standalone truss for which an analytical solution is available. The truss is located in an inclined position with respect to the 3D parallelepiped structure (Fig. 17). Displacement boundary conditions are specified to the nodes of the solid elements only. All nodes at the lower face are restricted against vertical movement. Perfect bond is assumed for the contact between truss members and solids. A concentrated force of 1 kN is directly applied to the top node of the truss. Since a very low Young modulus is considered for the solid material, it is expected that axial stresses in the truss elements are close to the analytical solution obtained considering the equilibrium of the sole frame.

The solid structure is discretized using non-structured tetrahedra, and the truss members are represented by the segments originated from the intersection between the tetrahedra elements and the truss frame. The mesh shown in Fig. 18 contains 256 four-node tetrahedra with five integration points, 22 linear rod elements with two integration points and 22 rod–solid interface elements with two integration points. The material parameters for this test are linear elastic and are presented in Table 3. Relatively high values for the interface stiffness parameters (k_s and k_n) are chosen to restrict the relative movement in the interface zone and hence model perfect bonding.

The analytical solution provides an axial stress value equal to $\sigma_s = \sqrt{2}/2$ MPa for the horizontal bar and $\sigma_s = -\sqrt{3}/2$ MPa for the two inclined bars. This fact allows computing the relative errors at the integration points of the rod elements. Fig. 19 presents an xz projection of the truss displaying perpendicular bars proportional to the relative errors at integration points. The numerical results match the analytical solution quite well with maximum relative errors below 3×10^{-4}. Since the analytical axial stress values are constant, linear elements produce satisfactory results. It is worth noting that the simulation behaves robustly especially considering that the discretization creates many short-length rod elements and that the (unsymmetrical) tetrahedral mesh is not an ideal setup for this simulation.

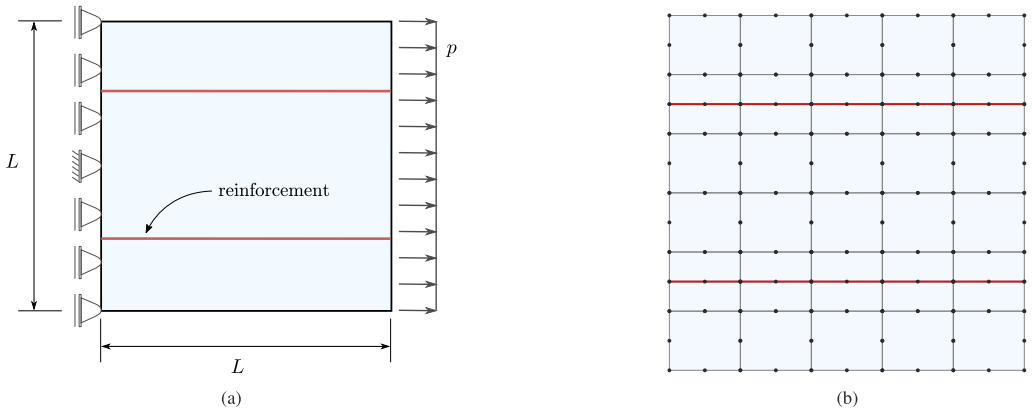


Fig. 15. Example3 — Square plate under tensile load. (a) test setup; (b) finite element mesh.

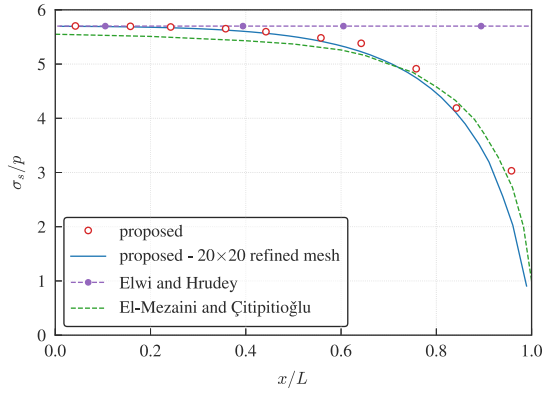


Fig. 16. Example 4 — Square plate under tensile load. Normalized axial stresses at the integration points of rod elements compared with the results using a refined mesh and the results from Elwi and Hrudey [13] and El-Mezaini and Çitipitioğlu [8].

Table 4
Example 5 — Pure bending of 3D reinforced beam. Material parameters.

Solids	$E = 20$ GPa	$\nu = 0.25$	
Bars	$E_s = 200$ GPa	$A_s = 0.001$ mM	
Rod–solid interfaces	$A = 0.001$ m^2	$k_s = 10^6$ GPa/m	$k_n = 10^6$ GPa/m

5.5. Example 5 — Pure bending of a 3D reinforced beam and mesh convergence test

A simply supported reinforced beam subjected to four-point bending is analysed in this section. First, the accuracy of the proposed method is demonstrated by comparing the stresses developed at the beam and along the rebars with results from an analytical solution. Subsequently, a mesh convergence analysis is performed to study the rate of convergence using the proposed formulation.

Fig. 20 shows the geometry of the reinforced concrete beam with the reinforcement layout and boundary conditions. The reinforcement is given by two straight rebars with 4 m length and 0.001 m^2 section area each. To better approximate the Bernoulli's beam theory, the supports are positioned at the beam's horizontal plane of symmetry. Fig. 21(a) shows the surface of the finite element mesh, and Fig. 21(b) shows the rebars. The mesh comprises 432 20-node quadratic hexahedral elements, 36 quadratic rod elements and 36 quadratic rod–solid interface elements, with 8, 2, and 2 integration points, respectively. The material properties for this study are linear elastic and are summarized in Table 4. The interface's contact properties are chosen to simulate perfect bonding between bars and the surrounding material.

To assess the numerical accuracy of this experiment, the stresses at the beam's top σ_{top} and bottom σ_{bot} and the reinforcement σ_s are compared with the analytical solution. The analytical solution is derived here by assuming the Bernoulli beam theory, elastic material behaviour and perfect bonding between the reinforcement and the surrounding solid material. Details of the analytical solution are contained in Appendix B.

The distributions of the numerical and analytical axial stresses along the beam and rebars are compared in Figs. 22(a), 22(b) and 23. The compressive and tensile stresses at the beam's top and bottom are computed at equally spaced points using interpolation from nodal values of solid elements, which in turn are computed from the integration points using the superconvergent patch recovery technique [41,42]. In turn, the stresses at reinforcements are taken directly from the integration points of rod elements. On the other hand, the stresses σ_{top}, σ_{bot} and σ_s obtained from the analytical solution are, as expected, constant in the region between the load points locations, in which the bending moment is uniform. An excellent agreement between numerical and analytical values is observed. The numerical stresses computed at $y = 1.4$ m and $y = 2.8$ m are higher in value due to stress concentrations near the load application points. Although these increments are not observed in the bar stresses, this can be explained by stress redistributions where the stresses at bulk and rod elements are compensated due to the equilibrium condition; thus the bending moment remains constant at the central part of the beam.

To assess the convergence properties in the proposed technique, an additional study is performed by running analyses with five meshes similar to the one presented in Fig. 21. These meshes are constructed by increasing the number of elements. They are composed of 8-node hexahedral elements, linear rod elements and linear rod–solid interface elements with 8, 2, and 2 integration points, respectively. The results obtained from five discretizations with degrees of freedom ranging from 510 to 20010 are presented in Fig. 24(a) that shows the distribution of the axial stresses σ_s in one rebar evaluated at the integration points. We observe that the rebar stresses converge consistently with the mesh refinement.

To further assess the convergence rate in this experiment, the axial stress at the centre of the rebar (middle length along the y direction) is compared against the closed-form solution. Fig. 24(b) plots the relative error on σ_s against the element size h. The relative error is defined as the difference between the numerical σ_s and the analytical value, divided by the latter. In turn, the element size h is defined as the diagonal distance of a solid element. As shown in the figure, quadratic convergence is verified. In practice, quadratic solid elements are recommended for this kind of simulation as it provides accurate results even with few elements, as shown in Fig. 23.

5.6. Example 6 — Pullout experiment

Pullout experiments are usually conducted to study the pullout strength of reinforced fibres and rebars embedded in concrete and

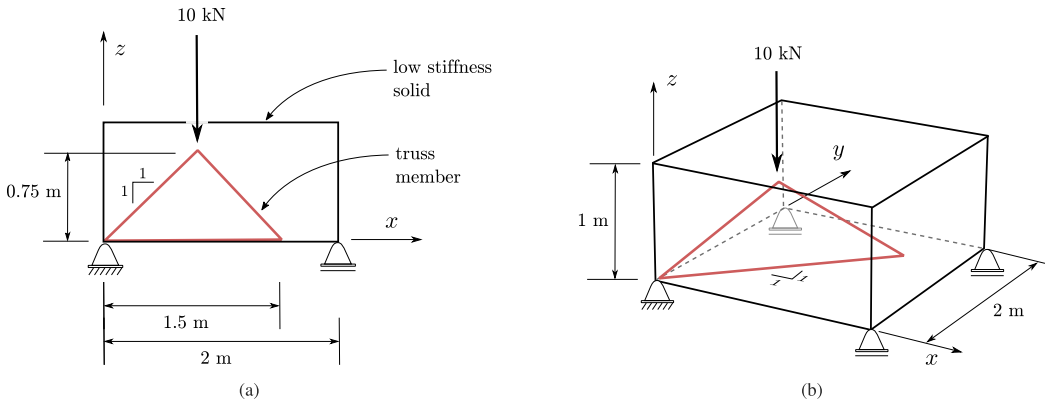


Fig. 17. Example 5 — Truss inside in a very soft material. Geometry and boundary conditions: (a) front view; (b) 3D view.

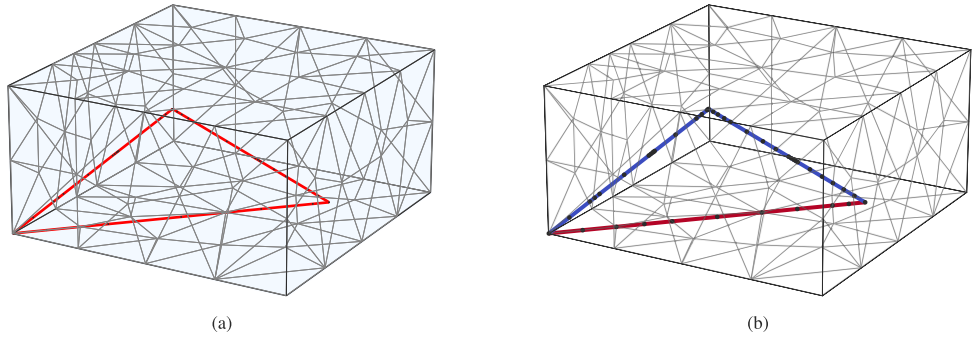


Fig. 18. Example 5 — Truss structured inside in a very soft volume: (a) surface of the finite element discretization; (b) internal rod elements.

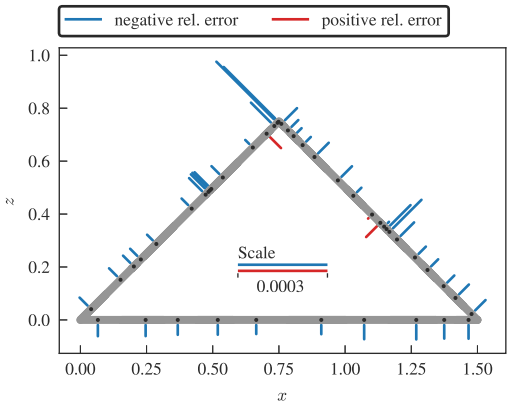


Fig. 19. Example 5 — Truss structure inside in a very soft material. xz projection showing relative errors on axial stresses at integration points.

Table 5
Example 6 — Pullout experiment. Material parameters.

Solids	$E = 30$ GPa	$\nu = 0.2$
Rods	$E_s = 200$ GPa	$\phi_s = 10$ mm
Rod–solid interfaces	$\tau_{max} = 9$ MPa	$\tau_{res} = 1$ MPa
	$s_{peak} = 0.55$ mm	$s_{res} = 8$ mm
	$k_s = 33$ GPa/m	$k_n = 10^3$ GPa/m
	$\alpha = 0.4$ mm	$\beta = 0.6$ mm

the contact properties in the interface between the two materials; see e.g. [43]. The simulation of a pullout experiment presented here aims to illustrate the process of total mobilization of the rod–solid interface's strength. The experiment is simulated by pulling the tip of a single reinforcing bar located inside a prismatic solid. The geometry and boundary conditions are illustrated in Fig. 25(a), and the material parameters are summarized in Table 5. The solid prismatic structure is fixed against movements at three faces, as indicated in the figure. A finite element mesh with 20-node hexahedra and 8 integration points is employed for the discretization (Fig. 25(b)). The rebars are simulated by quadratic rod elements with two integration points, and the rod–solid interface elements are quadratic as well, however, using three integration points. In the analysis, the pullout force is applied incrementally by means of a prescribed displacement $\delta = 9$ mm at the tip of the bar located at $y = 0.6$ m.

The slip distribution along the rebar is plotted in Fig. 26(a) for different loading levels of the prescribed displacement. As expected, the relative displacements are nonlinearly increasing along the rebar, with the maximum slip at the tip. Figs. 26(b) shows the distribution of the mobilized shear stress τ evaluated at the integration points of the rod–solid interface elements. It is observed that the shear strength is mobilized gradually, starting from the front tip of the bar and proceeding towards the interior as the imposed displacement is increased. Once a point at the interface reaches the contact strength (9 MPa), the shear stress decreases towards the residual contact strength (1 MPa). Fig. 27(a) shows the distribution of the axial rebar stress at different loading levels. The distributions show maximum values at the rebar's front tip and tend to zero at the interior tip.

Similarly to the shear stress distributions, the axial stresses increase in the first stages and decrease in the last ones. Fig. 27(b) shows the evolution of the pullout force according to the applied displacement δ at the rebar's front tip. After reaching the maximum pullout force (98.65 kN), the force decreases towards a residual level. For reference, the load–displacement curve from a case considering $\tau_{res} = \tau_{max} = 9$ MPa is also plotted in the figure. In both curves, the residual load values are compatible with the corresponding residual strength. For example, for the initial case with $\tau_{res} = 1$ MPa, the obtained residual

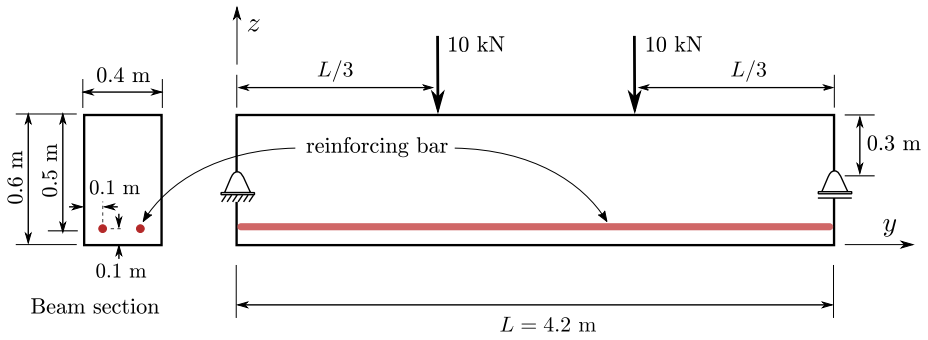


Fig. 20. Example 6 — Pure bending of 3D reinforced beam. Reinforcement layout and boundary conditions.

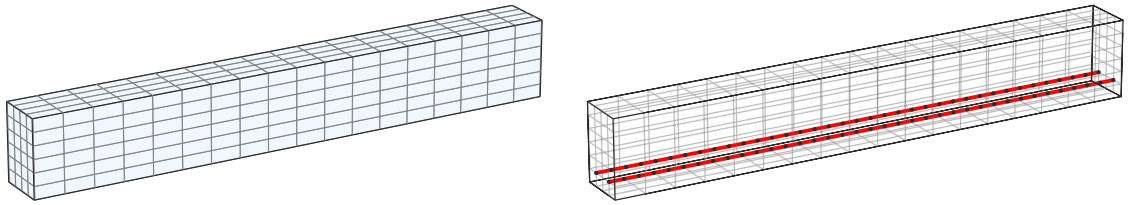


Fig. 21. Example 6 — Pure bending of 3D reinforced beam: (a) finite element mesh; (b) reinforcements discretization using quadratic elements.

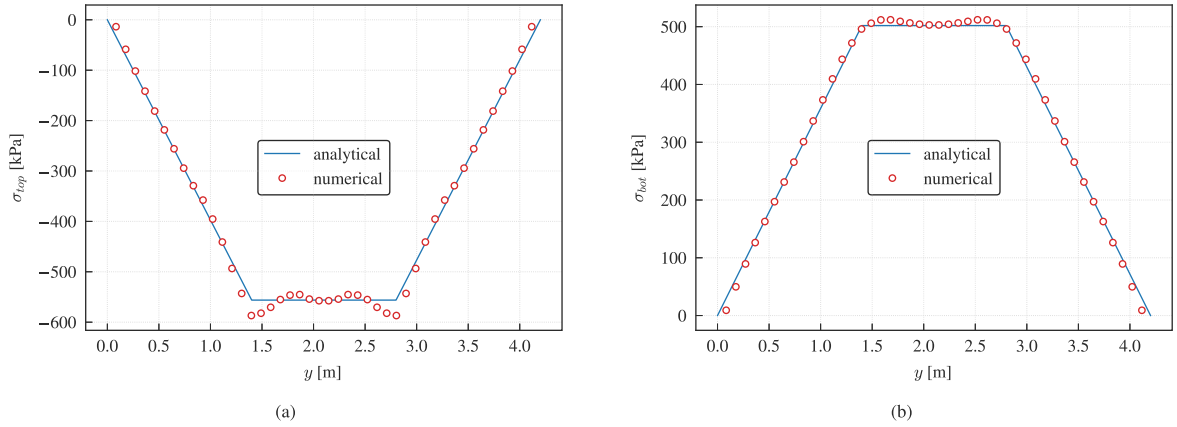


Fig. 22. Example 6 — Pure bending of 3D reinforced beam. Comparison between analytical and numerical stresses: (a) compressive stresses at the upper face; (b) tensile stresses at the bottom face.

force is $F_{res} = 12.51\,\text{kN}$, which is in good agreement with the calculated value based on the residual strength ($\pi\,\phi\,L\,\tau_{res} = 12.58\,\text{kN}$). As for the reference case with $\tau_{res} = 9\,\text{MPa}$, an excellent agreement can be observed between the numerical residual force $F_{res} = 113.1\,\text{kN}$ and the value computed from the residual strength (113.1 kN).

Finally, to assess the convergence rate, we run five analyses using finer meshes and observe the rebar stress and the contact stress distributions. The first fourth meshes are composed of 8-node hexahedral elements, linear rod elements and linear rod–solid interface elements. Within them, the number of degrees of freedom varies from 129 to 2388. The last mesh is composed of quadratic elements featuring 1227 degrees of freedom, and is used here as a reference. In this convergence study, we consider the results corresponding to an applied displacement equal to 2 mm. The results obtained at the integration points of rod and interface elements are presented in Fig. 28. According to Fig. 28(a), as the number of degrees of freedom increases, the rebar stress values tend to converge and get close to the reference curve. Likewise, Fig. 28(b) shows convergent results for the shear stresses at the interface. Based on these results, it is clear that quadratic elements should be preferred. Also, it is worth to recall that the axial stress is constant in linear rod elements and the bond stress is linear in linear interface elements.

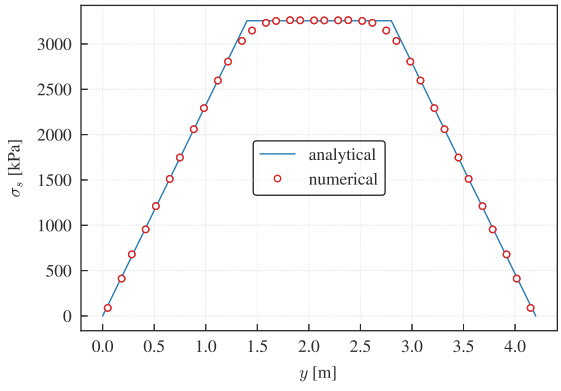


Fig. 23. Example 6 — Pure bending of 3D beam. Tension stress at integration points of rod elements.

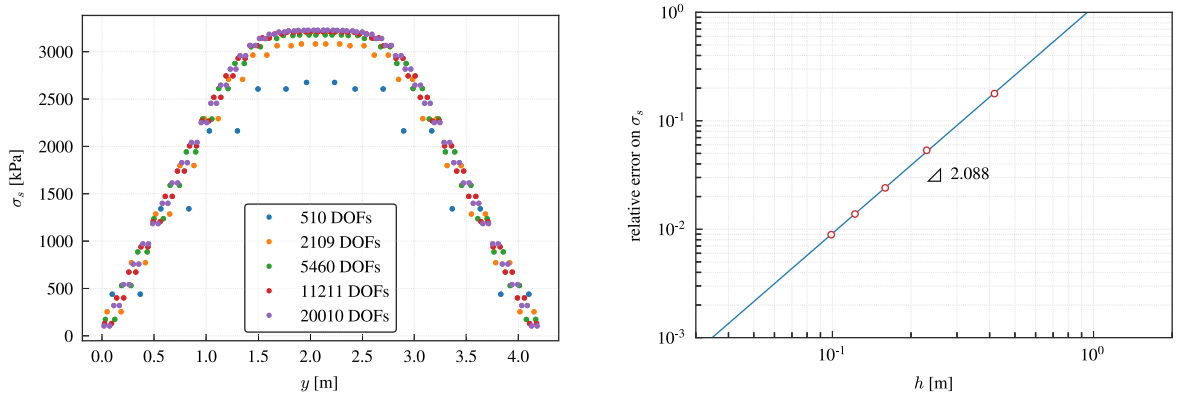


Fig. 24. Example 6 — Pure bending of 3D reinforced beam (convergence test): (a) axial stresses at one reinforcement bar evaluated at integration points for five different discretizations; (b) relative error of computed reinforcement stresses w.r.t. the analytical solution versus bulk elements size using 8-node hexahedra.

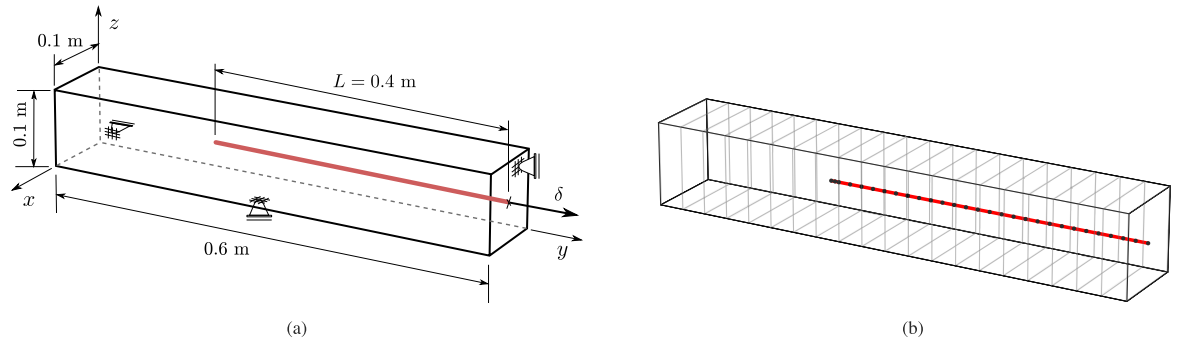


Fig. 25. Example 6 — Pullout experiment: (a) test setup and boundary conditions; (b) finite element discretization.

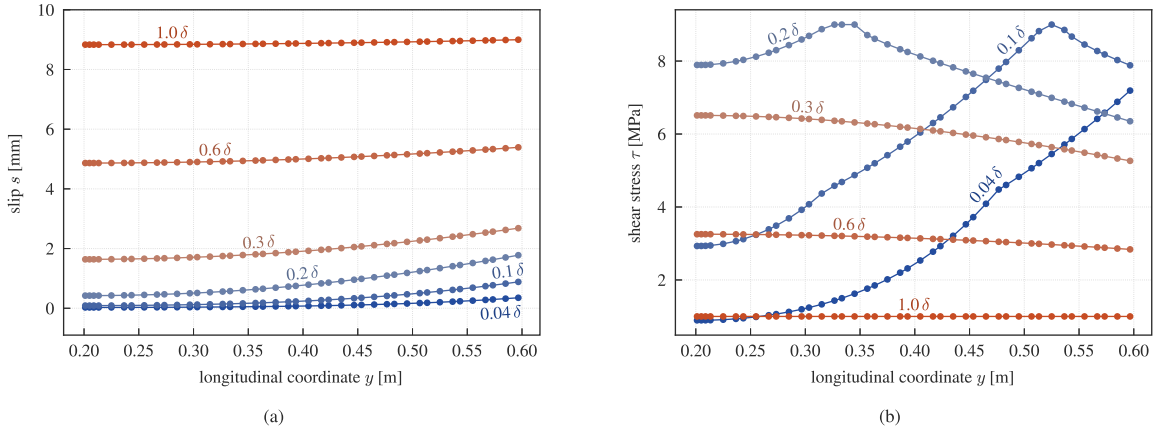


Fig. 26. Example 6 — Pullout experiment. Numerical results along the rebar computed at integration points of the interface element for different levels of the prescribed displacement: (a) rebar slip; (b) shear stress.

5.7. Example 7 — Cube stiffened by a spiral

This example presents a study on the effect of contact parameters for the rod–solid interface in a complex geometric setup. Furthermore, the example also shows the application of curved reinforcement. In this case, a cuboid structure with an internal spiral (Fig. 29) is simulated. The cube, with side lengths $1 \times 1 \times 1$ m, is fully fixed at its base. At the top face, a stiff plate (0.05 m thickness) is attached to transfer a prescribed vertical displacement to the combined solid-spiral material. Fig. 29(a) shows the finite element mesh surface, whereas Fig. 29(b) shows the spiral discretized in rod elements. The spiral geometry is given in parametric coordinates according to the following equations:

$$x = \frac{3}{8}\cos(t), \quad y = \frac{3}{8}\sin(t) \quad \text{and} \quad z = \frac{t}{8\pi}, \tag{55}$$

where $0 \le t \le 8\pi$. The whole mesh comprises 252 quadratic hexahedron elements, 82 quadratic rod elements and 82 quadratic rod–solid interface elements with 8, 2 and 3 integration points, respectively. Initially, no tip bearing elements were used.

The cube is loaded by applying vertical displacements (Δu_z = 25 mm) onto the top loading plate. Linear elastic material laws are adopted for the cube, the top plate and the spiral (Table 6). A set of numerical analyses are conducted to study the effect of modifying the

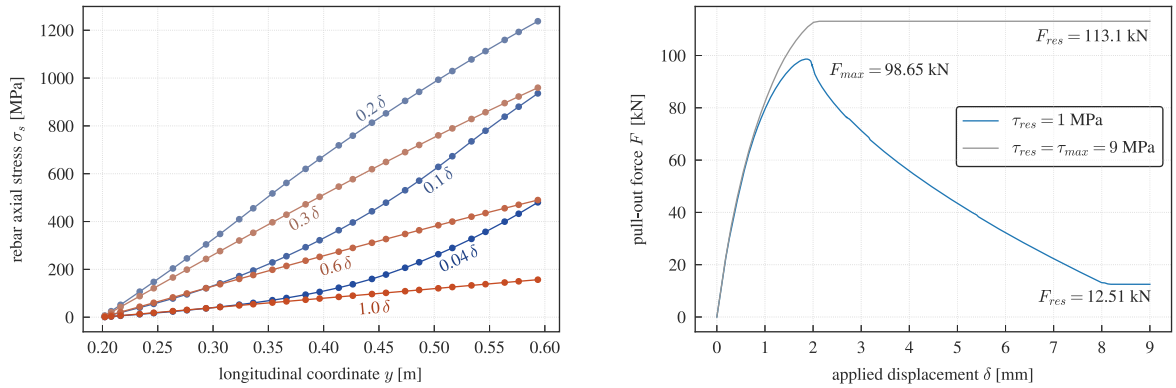


Fig. 27. Example 6 — Pullout experiment: (a) Axial stress along the rebar for different levels of the prescribed displacement; (b) Pullout force versus applied displacement.

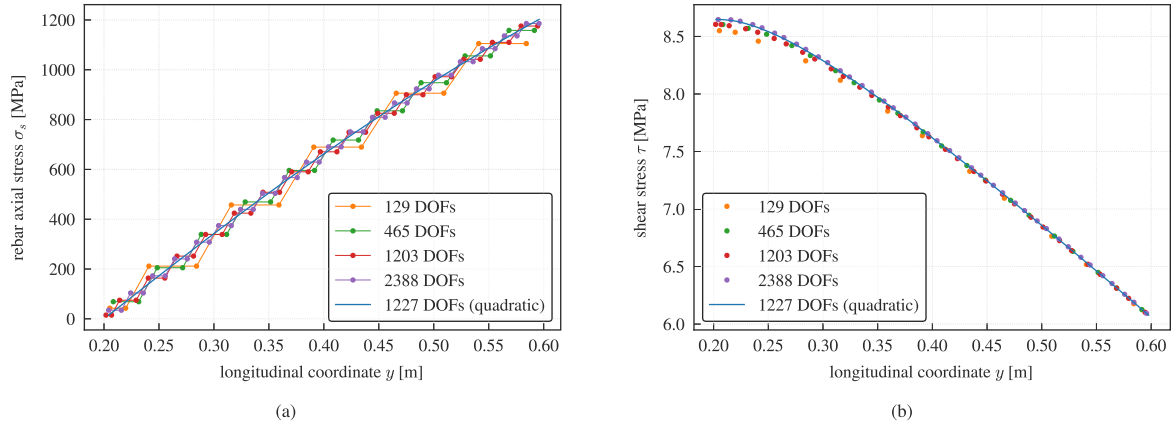


Fig. 28. Example 6 — Pullout experiment (convergence test). Results for different discretizations for pullout displacement equal to 2 mm: (a) axial stress at integration points of rod elements; (b) shear stress at integration points of the interface elements.

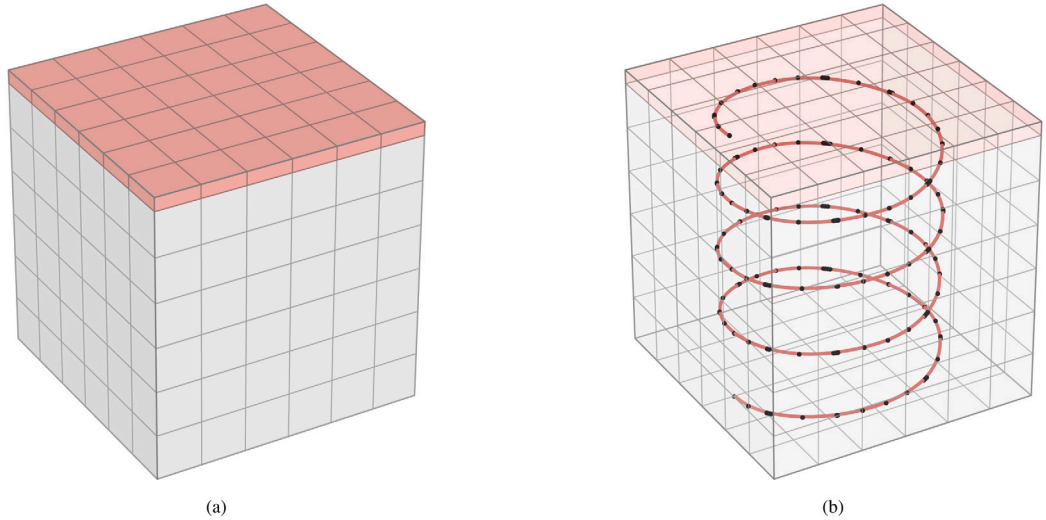


Fig. 29. Example 7 — Cube stiffened by a spiral: (a) surface of the finite element mesh highlighting the top plate elements; (b) rod elements for the reinforcing spiral.

interface constitutive model and contact parameters. These analyses and corresponding interface parameters are summarized in Table 7.

Fig. 30(a) shows the load–displacement results obtained at the plate for each analysis. The curve from Analysis S0 (unreinforced case) represents the lower bond of the load displacements curves. Conversely, the curve from Analysis S1 (linear interface) represents the upper bond where the composite presents the stiffest behaviour. The curve from Analysis S2 ($k_s \approx 0$) is close to the unreinforced case since the contact does not develop shear stresses, and the global stiffness is increased only due to the mechanical locking between the solid and the spiral. Regarding the nonlinear analyses, the curve from Analysis S3 ($\tau_{res} = 0$) shows a rapid load reduction after the complete interface failure; then, it matches the curve for $k_s \approx 0$. This result is expected since, once the interface fails, the condition $k_s \approx 0$ is similar to the condition $\tau =$

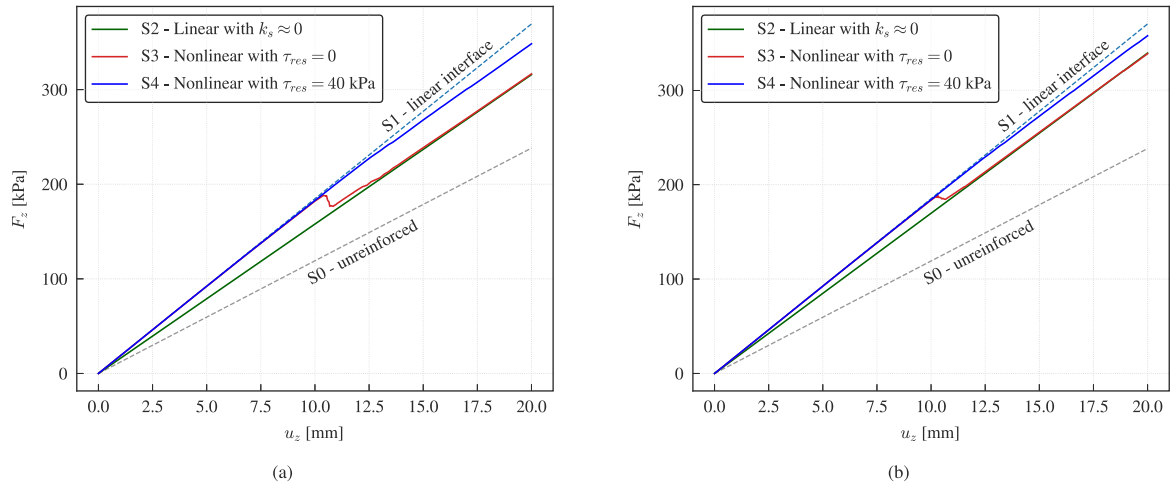


Fig. 30. Example 7 — Cube stiffened by a spiral. Force versus relative displacement curves for each analysis: (a) no bearing tips were considered; (b) considering the top endpoint as anchored.

Table 6
Example 7 — Cube stiffened by a spiral. Material properties for elastic materials (artificial values).

Cube	$E = 10$ MPa	$\nu = 0.45$
Top plate	$E = 200$ GPa	$\nu = 0.2$
Spiral	$E_s = 200$ GPa	$A_s = 0.005$ m^2
Tip contact (anchored, when used)	$k_{tip} = 10^7$ kN/m	

$\tau_{res} = 0$. On the other hand, the curve from Analysis S4 ($\tau_{res} = 40$ MPa) shows a slight loss of stiffness, implying that the interface elements fail gradually. Once all the interface reaches the residual stress, the curve becomes parallel to the one from Analysis S2. This result is also expected since, once the whole contact fails, the interface elements do not contribute to the global stiffness in the direction aligned with the reinforcement. The results from these analyses are useful to observe the mechanical effect of the reinforcement and the different outcomes obtained by changing the contact properties. For illustrative purposes, graphical results obtained at the end of Analysis S4 are presented. Thus, Fig. 31 shows the deformed mesh, whereas Figs. 32(a) and 32(b) present, respectively, the spiral axial stress and the corresponding slip. As expected, the stress and slip distributions are consistent with the loading and deformation.

Since the spiral is subjected to tension, no tip bearing elements were required at the spiral endpoints. However, to check the effect of a bearing tip, a second set of analyses were performed considering the spiral top end as an anchored tip. The parameter k_{tip} for the tip bearing element was considered equal to 10^7 kN/m. The results are presented in Fig. 30(b). Note that, in this case, the curves from Analyses S2, S3 and S4 are close to the linear interface curve. This shows the increment in the system stiffness due to the additional constraint. Fig. 33(a) shows the spiral axial stress. Different to the analysis with no anchored tip, a non-zero axial stress is observed at the top endpoint. Also, Fig. 33(b) shows no slip at this point.

5.8. Example 8 — Cyclic pullout test

This example reproduces the force–displacement curve of a cyclic pullout experiment presented by Murcia-Delso et al. [34]. Fig. 34(a) shows the experiment setup. The specimen consists of a 914 mm diameter concrete cylinder with a height of 15 times the nominal bar diameter $\phi = 43$ mm. The bar was bonded only in the central region over a length of 5ϕ. According to the authors, the short embedment length was intended to provide a fairly uniform bond-stress distribution at the contact and prevent yielding of the steel. Table 8 presents

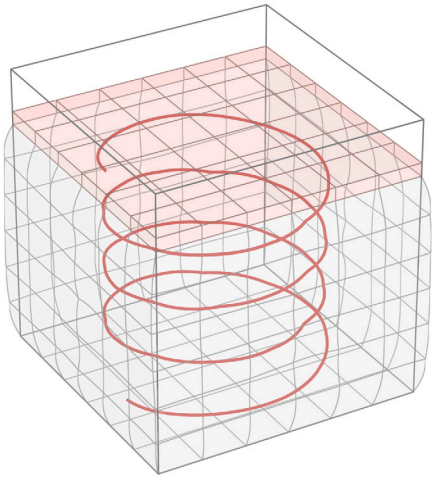


Fig. 31. Example 7 — Cube stiffened by a spiral. Deformation with 8× magnification at the end of Analysis S4.

the material parameters used in the simulation. Since the nonlinear behaviour is concentrated at the interface, linear elastic models are used for the concrete and steel. On the other hand, the proposed bond–slip model is used for the steel–concrete interface. For the unbounded region, the interface is assumed linear elastic with reduced stiffness in the longitudinal direction. The finite element mesh is presented in Fig. 34(b) and is composed by 225 quadratic hexahedron elements, 5 quadratic rod elements and 5 rod–solid interface elements. Although few interface elements were used in the region of interest, they were enough to reproduce the desired behaviour.

In the simulation, the top and bottom faces were considered fixed. Positive and negative displacements were applied to the rebar (Fig. 34(a)). Also, the reaction forces were logged at this location. Fig. 35 shows the numerical pull force compared to experimental results presented by Murcia-Delso and Benson Shing [44]. Good agreement can be observed between the numerical and experimental cyclic curves.

5.9. Example 9 — Reinforced concrete corbel

This example is designed to demonstrate the application of the proposed method by simulating an RC member. Putri [45] performed loading tests on four RC columns featuring a corbel. For this example,

Table 7
Example 7 — Cube stiffened by a spiral. Material parameters for each interface analysis (artificial values).

Analysis	Interface model	Parameters	
S0	None (unreinforced)	–	
S1	Linear	$k_s = 1$ GPa/m	$k_n = 10^3$ GPa/m
S2	Linear with no shear stiffness	$k_s \approx 0$	$k_n = 10^3$ GPa/m
S3	Nonlinear with no residual strength	$k_s = 1$ GPa/m	$k_n = 10^3$ GPa/m
		$\tau_{max} = 80$ kPa	$\tau_{res} = 0$
S4	Nonlinear with residual strength	$k_s = 1$ GPa/m	$k_n = 10^3$ GPa/m
		$\tau_{max} = 80$ kPa	$\tau_{res} = 40$ kPa
–	Remaining parameters for	$s_{peak} = 0.1$ mm	$s_{res} = 2$ mm
	Analyses S3 and s4	$\alpha = 0.4$	$\beta = 1.0$

Table 8
Example 8 — Cyclic pullout test. Constitutive models and material parameters.

Material/interface	Constitutive model	Parameters	
Concrete	Linear elastic	$E = 34.86$ GPa	$\nu = 0.2$
Steel	Linear elastic	$E_s = 210$ GPa	
Steel–concrete interface	Proposed nonlinear $\tau^0_{max} = 23.8$ MPa	$\tau^0_{res} = 5.95$ MPa	
		$s^0_{peak} = 3.0$ mm	$s_{res} = 24.9$ mm
		$\alpha = 0.5$	$\beta = 0.85$
		$k_s = 5 \times 10^7$ kPa/m	$k_n = 10^6$ GPa/m
Unbounded interface	Linear elastic	$k_s = 1$ kPa/m	$k_n = 10^6$ GPa/m

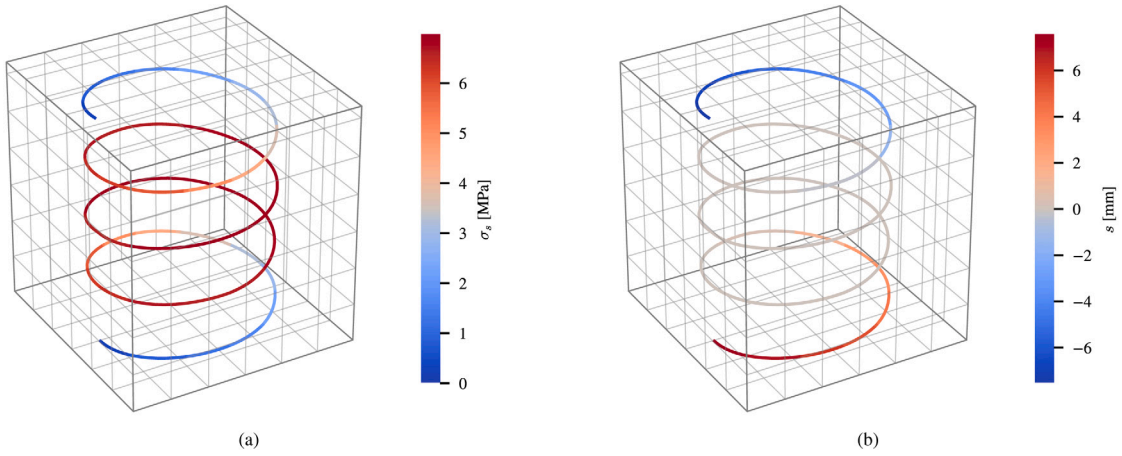


Fig. 32. Example 7 — Cube stiffened by a spiral. Results at the end of Analysis S4: (a) axial strength at the reinforcement; (b) reinforcement slip.

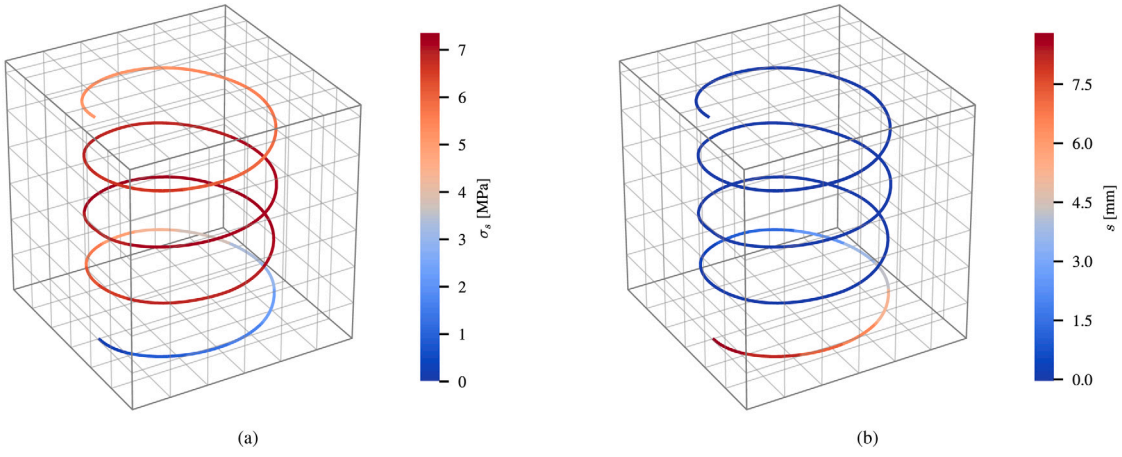


Fig. 33. Example 7 — Cube stiffened by a spiral. Results at the end of Analysis S4 considering the top spiral end as anchored: (a) axial strength at the reinforcement; (b) reinforcement slip.

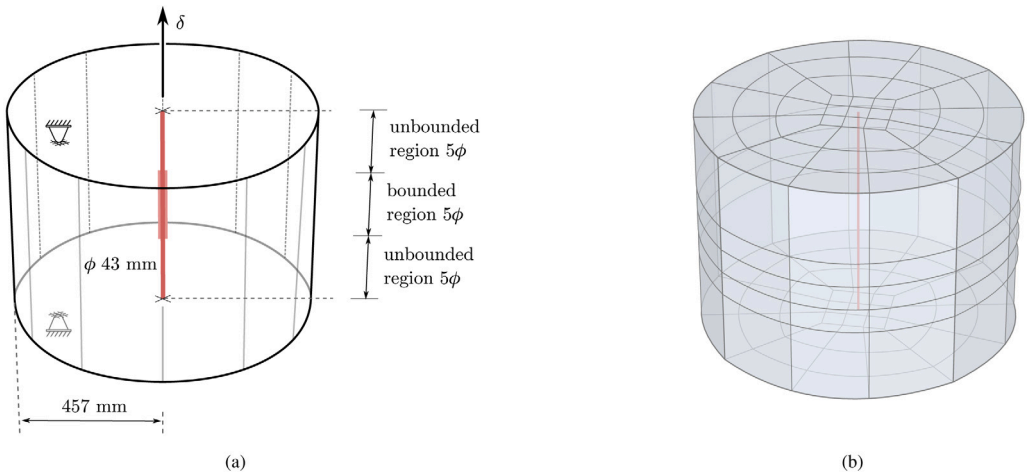


Fig. 34. Example 7 — Cyclic pullout test: (a) test setup; (b) surface of the finite element mesh.

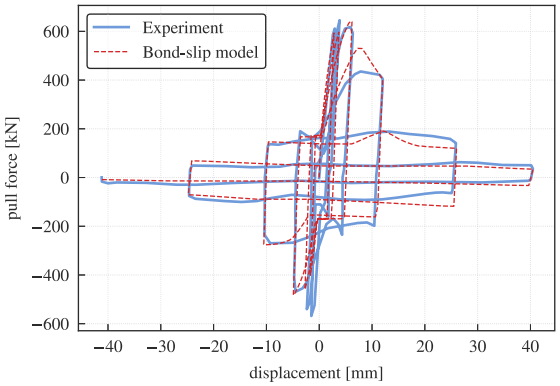


Fig. 35. Example 8 — Cyclic pullout test. Numerical prediction compared to experimental results presented by Murcia-Delso and Benson Shing [44].

we selected the specimen KpSNI-02 for numerical simulation. Fig. 36 displays the geometry, steel reinforcement layout and boundary conditions. The corbel's primary reinforcement is given by two 13 mm and two 10 mm rebars. In addition, three 6 mm horizontal stirrups, spaced at 80 mm intervals, are employed to reinforce the corbel. In turn, the column is reinforced by ten 13 mm longitudinal rebars and 8 mm ties spaced at 150 mm intervals. A bearing plate 60 mm wide was considered for the load application. Fig. 37(a) shows the surface of the finite element mesh, and Fig. 37(b) shows a 3D view of the rebars' layout used in the simulation.

The finite element mesh comprises 4-node tetrahedra, quadratic rods and quadratic rod–solid interface elements with 4, 2 and 3 integration points, respectively. Also, tip bearing elements (non anchored) were placed at the ends of the corbel's primary reinforcement. All parameters are summarized in Table 9. The rebars are considered elastic-perfectly plastic, while the rod–solid interfaces are simulated using the proposed bond–slip model. As for the concrete, we employed a discrete crack model that uses cohesive elements (see [40,46]) in conjunction with the compression stress–strain relationship proposed by Carreira and Chu [47]. All required steel parameters are available in the experimental work, including the yielding stress σ_y. The concrete compressive strength was reported as $f_c = 31$ MPa, and remaining parameters for the modelling were adopted accordingly. In Table 9, f_c and f_t are, respectively, the concrete compressive and tensile strengths, E_{c0} is the initial tangent elastic modulus, $\epsilon_{c,peak}$ is the strain corresponding with the compressive strength, G_F stands for the concrete fracture energy and $\tan(\varphi)$ for the tangent of the concrete friction angle. As for

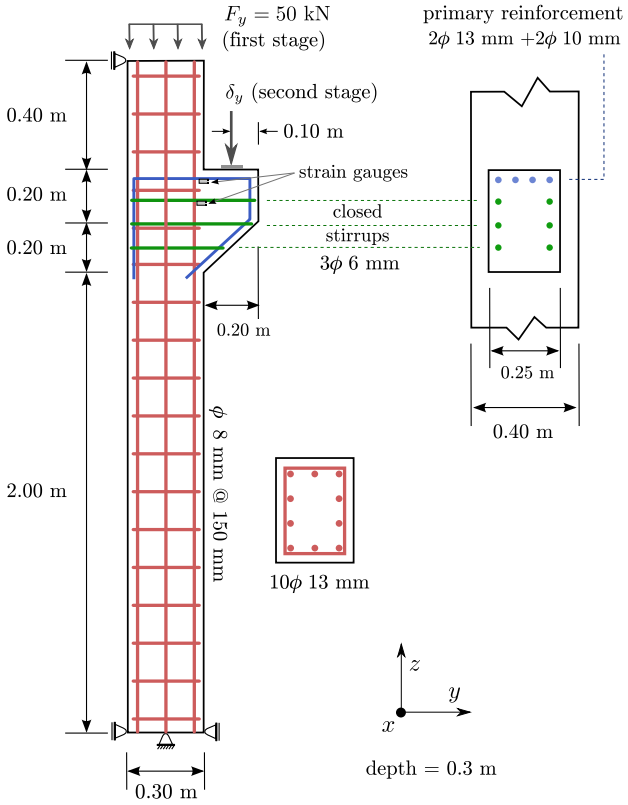


Fig. 36. Example 9 — Reinforced concrete corbel. Geometry and boundary conditions.

the interface, the parameters were estimated according to the rebar diameter and the approximation formulas described in [34], such as $s_{peak} = 0.07\phi$, $s_{res} = 0.5\ell_{rib}$ and $\tau_{res} = 0.25\tau_{max}$, where ℓ_{rib} is the rebar rib spacing.

The analysis is carried out in two stages, analogous to the experiment. The first stage considers the application of a 50 kN load at the column top surface. In the second stage, an increasing load F_z is applied to the corbel until failure. In the simulation, the load was applied by small increments of prescribed displacement until failure. During the experiment, strain gauges were fixed on the primary reinforcement (13 and 10 mm rebars) and the upper corbel stirrup (6 mm rebar) at locations that match the column–corbel junction, as shown in Fig. 36. In the numerical analysis, the rebar strains at the nearest integration points to

Table 9
Example 9 — Reinforced concrete corbel. Constitutive models and material parameters.

Material/interface	Constitutive model	Parameters			
Concrete	Discrete crack	$E = 20$ GPa	$\nu = 0.2$	$f_c = 31$ MPa	$\varepsilon_{c,peak} = 0.003$
		$f_t = 2.0$ MPa	$G_F = 70$ N/m	$\tan(\varphi) = 1.1$	
Bearing plate	Elastic-perfectly plastic	$E = 200$ GPa	$\nu = 0.3$		
Steel ϕ 6 mm	Elastic-perfectly plastic	$E_s = 180$ GPa	$\sigma_y = 340$ MPa		
Steel ϕ 8 mm	Elastic-perfectly plastic	$E_s = 200$ GPa	$\sigma_y = 500$ MPa		
Steel ϕ 10 mm	Elastic-perfectly plastic	$E_s = 200$ GPa	$\sigma_y = 490$ MPa		
Steel ϕ 13 mm	Elastic-perfectly plastic	$E_s = 200$ GPa	$\sigma_y = 460$ MPa		
Steel–concrete	Proposed	$\tau_{max} = 12$ MPa	$\tau_{res} = 3$ MPa	$k_s = 20$ GPa/m	$k_n = 10^6$ GPa/m
interface ϕ 13 mm		$s_{peak} = 0.9$ mm	$s_{res} = 3$ mm	$\alpha = 0.4$	$\beta = 0.8$
Steel–concrete	Proposed	$\tau_{max} = 10$ MPa	$\tau_{res} = 2.5$ MPa	$k_s = 20$ GPa/m	$k_n = 10^6$ GPa/m
interface ϕ 16 mm		$s_{peak} = 0.7$ mm	$s_{res} = 2.5$ mm	$\alpha = 0.4$	$\beta = 0.8$
Steel–concrete	Proposed	$\tau_{max} = 7$ MPa	$\tau_{res} = 1.75$ MPa	$k_s = 20$ GPa/m	$k_n = 10^6$ GPa/m
interface ϕ 8 mm		$s_{peak} = 0.55$ mm	$s_{res} = 2$ mm	$\alpha = 0.4$	$\beta = 0.8$
Steel–concrete	Proposed	$\tau_{max} = 5$ MPa	$\tau_{res} = 1.25$ MPa	$k_s = 20$ GPa/m	$k_n = 10^6$ GPa/m
interface ϕ 6 mm		$s_{peak} = 0.4$ mm	$s_{res} = 1.5$ mm	$\alpha = 0.4$	$\beta = 0.8$
Tip contact	Proposed (non anchored)	$k_{tip} = 10^7$ kN/m			

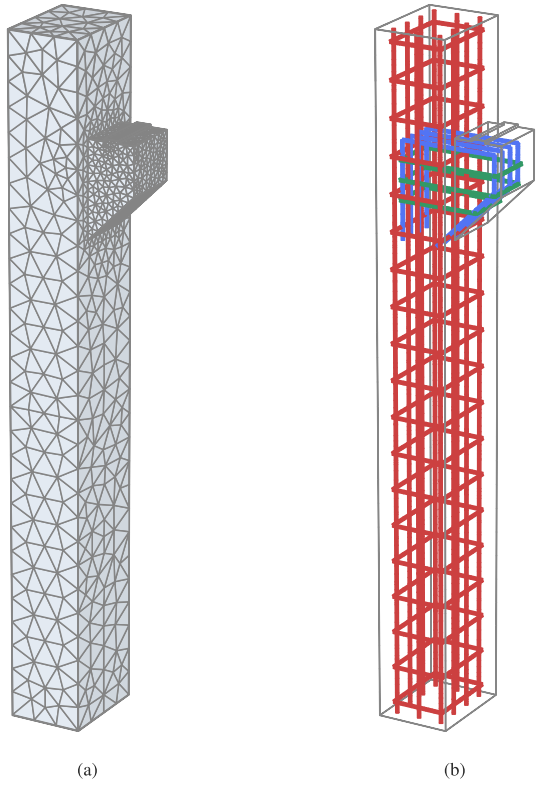


Fig. 37. Example 9 — Reinforced concrete corbel: (a) surface of the finite element mesh; (b) reinforcement layout.

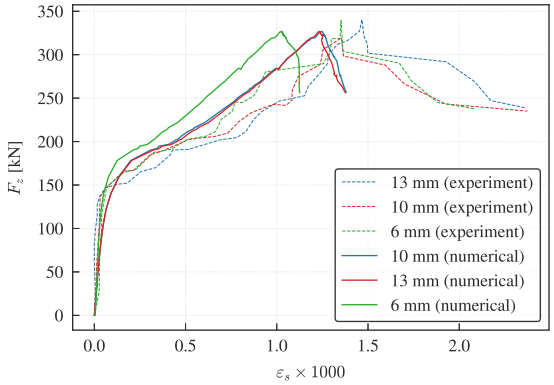


Fig. 38. Example 9 — Reinforced concrete corbel. Comparison between numerical and experimental results at rebar strain gauges.

these locations are logged for comparison with the experimental values. The results are shown in Fig. 38 where we can observe fair agreement between the experimental and numerical strain curves. Also, the peak load is well approximated. In the simulation, the upper 6 mm stirrup plastified near the peak load; however, at a different location than the corresponding strain gauge. This is compatible with the observations reported in the experimental study. It is important to stress that the simulations of RC members involve essential aspects other than the reinforcement and interface modelling, for example, the definition of proper boundary conditions according to the experiment and the choice of constitutive model and proper concrete parameters.

Fig. 39 presents graphical results when the corbel load is equal to $F_z = 300$ kN. Fig. 39(a) shows the distribution of axial stress in the steel reinforcement. The rebars coloured with red shades are in tension, while the rebars coloured with blue shades are in compression. The maximum tensile stresses are found in the horizontal portions of the corbel primary reinforcements. In contrast, the maximum compressive stresses are found in the longitudinal rebars at locations near the lower part of the corbel. Fig. 39(b) presents a close view of the corbel showing the resulting crack patterns using a magnification factor of 30. As can be seen, the cracks develop starting from the bearing plate.

To further investigate the stress at the corbel reinforcement, Fig. 40 plots the axial stress at the integration points of 13 mm rebar for two load levels. Thus, Fig. 40(a) shows the axial stresses for $F_z = 150$ kN, where the corbel is mostly at the elastic regime. As expected, the horizontal portion of the rebar is subjected to tensile stress (blue), while the inclined portion is under compression (red). On the other hand, Fig. 40(b) shows the stresses $F_z = 300$ kN. At this point, the tensile stress at the horizontal portion of the rebar is much higher but not enough to reach the steel yielding stress. In both cases, it is interesting to note that the stress distributions are almost constant along the compressed rebar segments. Similarly to the axial stress, Fig. 41(a) shows the corresponding bond stress obtained at integration points of the interface for $F_z = 150$ kN. Note that there is some interference in the stress distribution near the bearing plate location and the corbel–column junction. Fig. 41(b) shows the shear stress $F_z = 300$ kN. In both cases, the bond stress distribution at the horizontal rebar clearly shows an inversion near the load location. Also, due to the use of bearing tip elements, note that the bond stress at the rebar ends tend to zero, which corresponds to the absence of relative displacements at these locations.

6. Conclusions

This paper presented a complete formulation for modelling reinforced solids (2D or 3D; bulk) with bars (2D or 3D lines, curved or

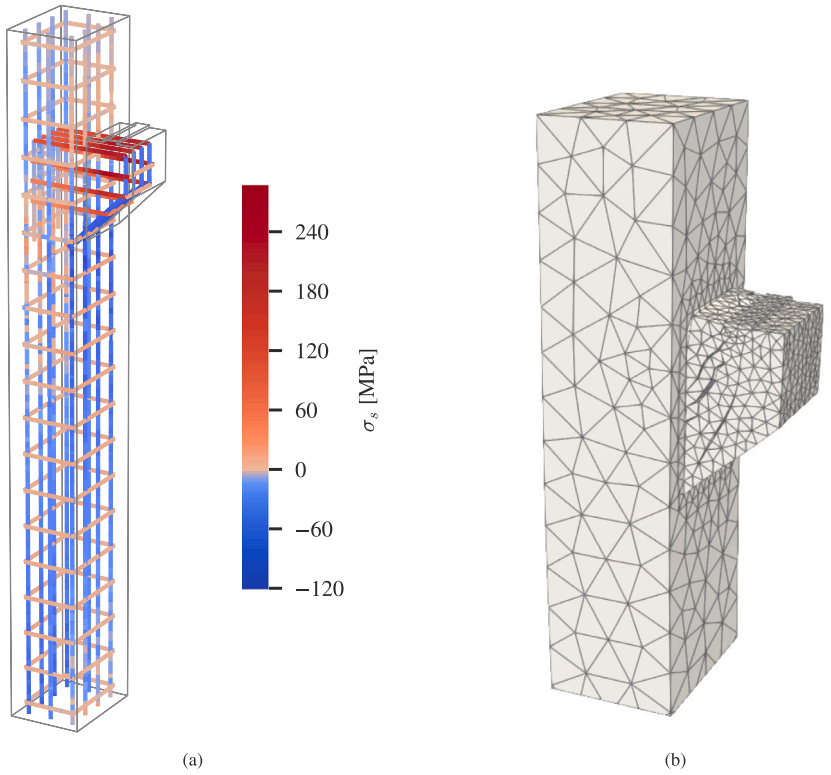


Fig. 39. Example 9 — Reinforced concrete corbel. Graphical results for $F_z = 300$ kN: (a) Axial stress at rod elements. (b) Close-up of the corbel displaying the crack patterns using a magnification factor of 15.

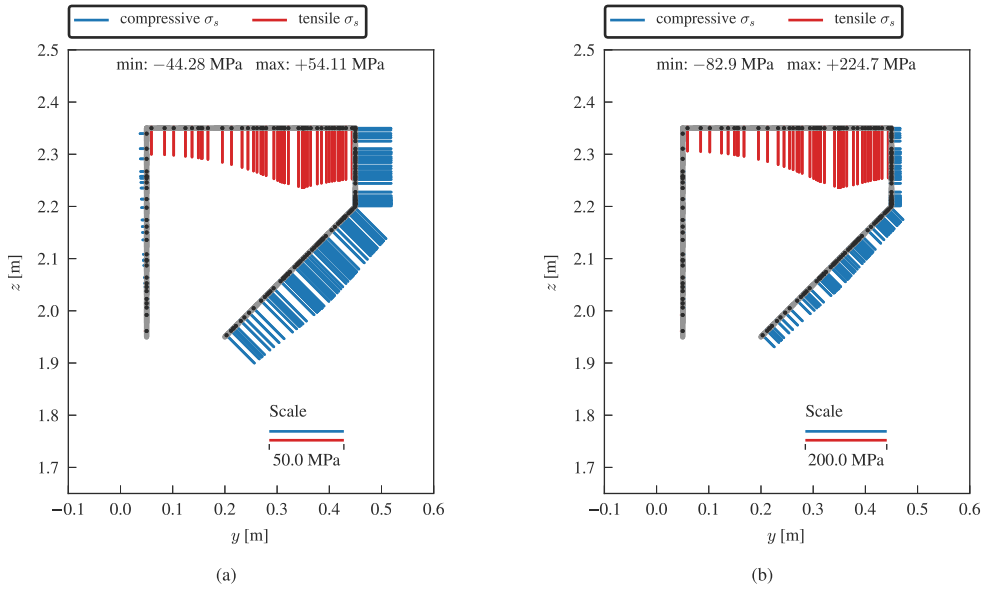


Fig. 40. Example 9 — Reinforced concrete corbel. Axial stresses at integration points of the selected corbel rebar: (a) results at 0.5 mm of the applied displacement; (b) results at the end of the analysis.

not) using finite elements. The proposed methodology is practical and convenient and can represent lines arbitrarily located inside a bulk finite element mesh without the need for nodal compatibility. To this end, a novel continuous rod–solid interface element is formulated to link bulk and line elements and model relative displacements and bond stresses. The interpolation functions for the interface element are the same as the linked rod element. The element stiffness matrix and internal forces are computed using numerical integration. The number of integration points of an interface element can differ from those used in the corresponding rod element. We recommend the use of two integration points for linear interpolation and three integration points for quadratic interpolation. This interface element is different to spring-like elements since the equations are derived from the equilibrium condition along the rod–solid contact and not at single points. This feature allows finding bond stress values at the integration points of the interface element, similarly to conventional finite elements. Those values can be recovered at nodal points and later interpolated to obtain the bond stress distribution along the rod. By varying the interface

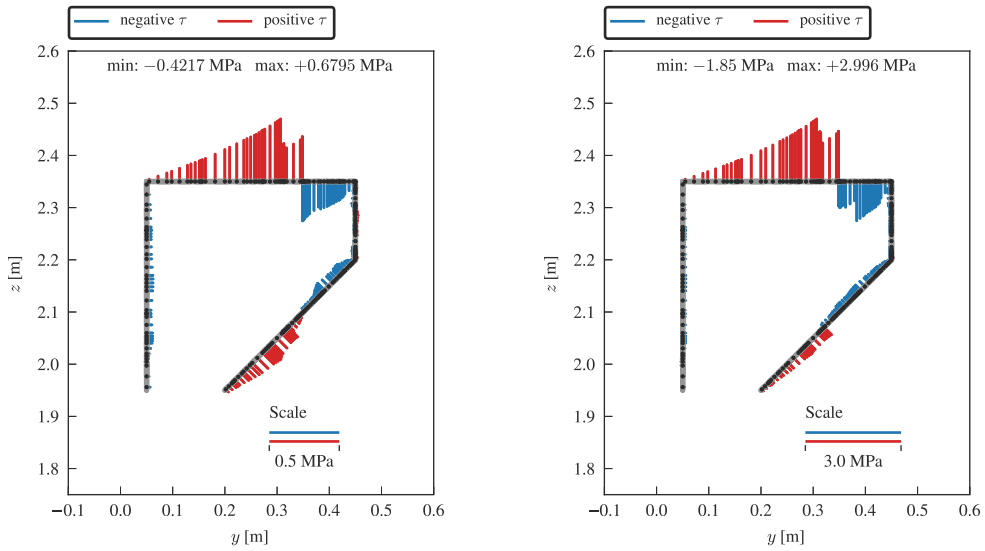


Fig. 41. Example 9 — Reinforced concrete corbel. Shear stresses at integration points of interface elements corresponding to the selected corbel rebar: (a) results at 0.5 mm of the applied displacement; (b) results at the end of the analysis.

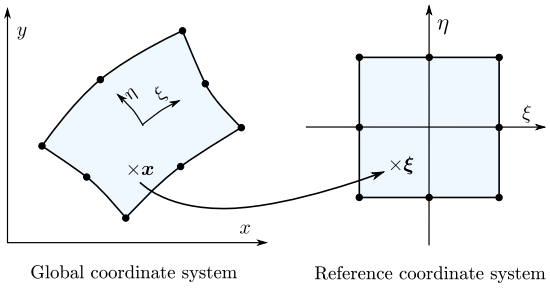


Fig. A.42. Inverse mapping representation.

properties along the rebar, it is possible to simulate perfect bonding zones and free sliding zones. An arbitrary bond–slip model can be used in conjunction with the interface element. For the sake of completeness, a cyclic bond–slip model was presented. Unlike existing embedded approaches, the proposed method allows for the direct application of boundary conditions at rebar nodes, such as forces and displacements. Also, a tip bearing element is formulated to model interaction between the reinforcement endpoints and the surrounding solid.

To fully demonstrate the capabilities of the proposed methodology, several numerical analyses are presented. The first analyses deal with patch tests and comparisons with closed-form solutions where very accurate results are obtained. Two convergence tests are also presented. It was shown that the formulation exhibits quadratic convergence whenever theoretically possible. The bond–slip model was tested by simulating monotonic and cyclic pullout tests showing satisfactory results. The last analyses deal with more complex situations such as an elastic cube reinforced by a spiral and a reinforced concrete member. In these simulations, the axial and bond stresses along the reinforcements were predicted. Also, the proposed tip bearing element was used to simulate anchored and non-anchored tips successfully. Furthermore, two experimental tests were simulated. Good agreement was observed between numerical and experimental data.

In summary, the proposed methodology can be conveniently applied to the study of complex RC structures or any solid medium reinforced by fibres. The following limitations should be noted nonetheless: (a) the presented formulation does not consider the flexural resistance of the reinforcing bars, which may be necessary in specific applications and can be incorporated by using beam elements instead of rod elements; (b) since the reinforcements are assumed as line elements, the proposed formulation cannot accurately model the interactions of large-diameter bars with their surrounding medium.

CRediT authorship contribution statement

R. Durand: Conceptualization, Methodology, Resources, Software, Validation, Writing – original draft, Writing – review & editing. **M.M. Farias:** Conceptualization, Resources, Reviewing. **D.M. Pedroso:** Resources, Software, Writing – review & editing. **G. Meschke:** Reviewing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Inverse mapping

Given a bulk element, the process of finding the reference coordinates ξ of a point in the global coordinate system $\boldsymbol{x}$, as illustrated in Fig. A.42, is known as *inverse mapping*, see, e.g. [13,14,32]. In other than linear elements, this process leads to a nonlinear system of equations. To solve this system, a Newton–Raphson scheme is adopted here due to the simplicity and quadratic convergence.

For an isoparametric element, the spatial coordinates, $\boldsymbol{x}$, are approximated using direct mapping as

$$\boldsymbol{x}(\xi) \approx \sum_m N_m(\xi)\, \boldsymbol{x}_m = \boldsymbol{X}\, \boldsymbol{N}(\xi), \tag{A.1}$$

where N_m are the interpolation (shape) functions of the element, $\boldsymbol{x}_m$ are the nodal coordinates of the element, $\boldsymbol{X}$ is a matrix whose columns

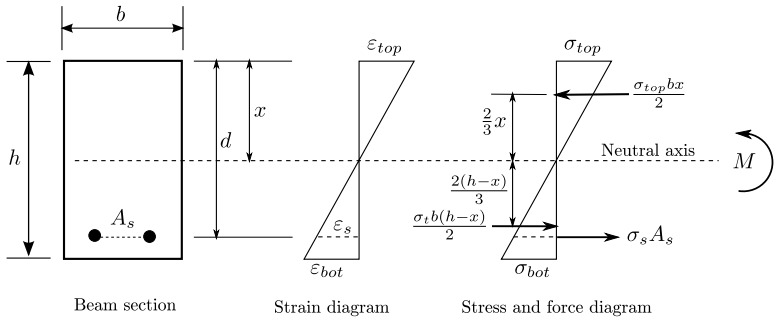


Fig. B.43. Pure bending of 3D reinforced beam. Stress and strain distributions for deriving an analytical solution.

are given by the nodal coordinates and $\boldsymbol{N}$ is a vector containing the values from nodal shape functions.

The inverse mapping algorithm starts by choosing a trial point $\xi = \xi_{tr}$ in the reference system corresponding to the spatial input point $\boldsymbol{x}_{in}$. The trial point can be, for example, the centre of the cell in the reference system. Then, by taking the difference between $\boldsymbol{x}_{in}$ and its isoparametric approximation (Eq. (A.1)), a residual can be computed as

$$\boldsymbol{r}(\xi) = \boldsymbol{x}_{in} - \boldsymbol{X}\,\boldsymbol{N}(\xi). \tag{A.2}$$

Approximating the residual by a truncated Taylor's series, the iterative Newton's method can be used to find ξ. Thus at iteration $k+1$, we have

$$\boldsymbol{r}_{k+1} \approx \boldsymbol{r}_k + \frac{d\boldsymbol{r}}{d\xi}\,\Delta\xi = 0, \tag{A.3}$$

where $\Delta\xi = \xi_{k+1} - \xi_k$. The derivative of the residual is a second-order tensor and can be computed by

$$\frac{d\boldsymbol{r}}{d\xi} = -\sum_m \boldsymbol{x}_m \frac{dN_m(\xi)}{d\xi} = -\boldsymbol{X}\boldsymbol{N}_D = -\boldsymbol{J}. \tag{A.4}$$

Therein, $\boldsymbol{N}_D$ is a matrix containing the derivatives of the shape functions evaluated at ξ and $\boldsymbol{J}$ is the bulk element's Jacobian of the direct mapping. Therefore, from Eq. (A.3)

$$\Delta\xi = \boldsymbol{J}^{-1}\boldsymbol{r}_k. \tag{A.5}$$

Thus, the reference coordinates at iteration $k+1$ are found as

$$\xi_{k+1} = \xi_k + \Delta\xi. \tag{A.6}$$

The process is repeated until the norm of $\Delta\xi$ is smaller than a specified tolerance. The complete process is summarized in Algorithm 3.

Algorithm 3: Inverse mapping.

Data: Bulk element data, point coordinates $\boldsymbol{x}_{in}$, trial reference coordinates ξ_{tr}, tolerance *tol*
Result: Reference coordinates ξ

$\xi = \xi_{tr}$ *! initial value*
do
  $\boldsymbol{N} = \boldsymbol{N}(\xi)$ *! shape functions vector*
  $\boldsymbol{N}_D = \boldsymbol{N}_D(\xi)$ *! shape functions derivatives*
  $\boldsymbol{J} = \boldsymbol{X}\boldsymbol{N}_D$ *! Jacobian matrix*
  $\boldsymbol{r} = \boldsymbol{x}_{in} - \boldsymbol{X}\boldsymbol{N}$ *! residual*
  $\Delta\xi = \boldsymbol{J}^{-1}\boldsymbol{r}$ *! reference coordinates increment*
  $\xi = \xi + \Delta\xi$ *! current ξ approximation*
until $||\Delta\xi|| < tol$

Appendix B. Analytical stresses for an elastic reinforced beam

For a linear elastic beam with longitudinal reinforcement, this analytical solution aims to provide the longitudinal stresses at the beam's top and bottom faces and at the reinforcement. The following assumptions are considered:

1. the solid and rebars behave as linear elastic materials;
2. flat cross-sections along the beam remain flat after deformation;
3. as a consequence of the previous assumption, σ_{yy} stresses and ε_{yy} strains along a particular section are linearly distributed;
4. there is perfect bonding between the beam and rebars; hence the beam and rebars have the same strain field.

Fig. B.43 shows a cross-section of the elastic beam where x is the neutral axis position, d is the effective depth of the beam, and A_s is the total cross-sectional area of the rebars. σ_{top} and σ_{bot} indicate the compressive and tensile stresses at the top and bottom face of the beam, respectively and ε_c and ε_t are the corresponding linearized strains. The stresses and strains in the steel reinforcement are denoted as σ_s and ε_s, respectively.

The analytical solution is obtained by considering the following equations

$$0 = \tfrac{1}{2}\,\sigma_{top}\,x\,b - \tfrac{1}{2}\,\sigma_{bot}\,(h-x)\,b - \sigma_s\,A_s, \tag{B.1}$$

$$M = \tfrac{1}{2}\,\sigma_{top}\,x\,b\left(d - \tfrac{x}{3}\right) - \tfrac{1}{2}\,\sigma_{bot}\,b\,(h-x)\left[\tfrac{1}{3}\,(h-x) - (h-d)\right], \tag{B.2}$$

$$x = \frac{\sigma_{top}\,h}{\sigma_{top} + \sigma_{bot}} \quad \text{and} \tag{B.3}$$

$$\sigma_s = \frac{E_s}{E}\,\sigma_{bot}\,\frac{d-x}{h-x}. \tag{B.4}$$

There, Eq. (B.1) represents the equilibrium of forces, Eq. (B.2) represents the equilibrium of bending moment, Eq. (B.3) represents the neutral axis position, and Eq. (B.4) originates from the strain compatibility assumption. Besides, E and E_s are the Young modulus from solids and rebars, respectively. With the bending moment M as input, for a given section, the solution of the system yields

$$x = \frac{b\,E\,h^2 + 2\,d\,A_s\,E_s}{2\,(b\,E\,h + A_s\,E_s)}, \tag{B.5}$$

$$\sigma_{top} = \frac{6\,M\,x}{\kappa\,b\,h}, \tag{B.6}$$

$$\sigma_{bot} = \frac{6\,M\,(h-x)}{\kappa\,b\,h} \quad \text{and} \tag{B.7}$$

$$\sigma_s = \frac{3\,M\,(2x-h)}{\kappa\,A_s}; \tag{B.8}$$

where the following auxiliary variable is defined

$$\kappa = -3\,h\,x - 3\,h\,d + 6\,x\,d + 2\,h^2 \tag{B.9}$$

Note that the solution above is different from the one found in the reinforced concrete literature; see, e.g. [48]. The reason is that our solution considers a fully elastic beam. In contrast, the conventional design, based on the ultimate limit state, neglects the tension strength due to the occurrence of cracks.

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